

FEDERAL SECURITY AGENCY
UNITED STATES PUBLIC HEALTH SERVICE
NATIONAL OFFICE OF VITAL STATISTICS



UNITED STATES LIFE TABLES
and
ACTUARIAL TABLES
1939-1941

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UNITED STATES LIFE TABLES AND ACTUARIAL TABLES, 1939-1941

PART I

INTRODUCTION

Plan and scope of this volume

The life tables in this volume are based on the 1940 census of population and the deaths of the 3-year period 1939-1941. Separate life tables have been prepared for each sex for each of three racial groups: white, Negro, and other races. This is the first time official life tables have been prepared for races other than whites and Negroes in the United States. Life tables are also included for the total population of each sex, for the total population of each racial group without distinction by sex, and for the entire population without distinction by race or sex. Each of the 12 life tables is based on data for the entire continental United States. Also included are certain actuarial tables derived from the life tables for white males, white females, and total whites, to be used in calculating premiums and values for life annuities, life assurances, and other monetary benefits contingent on death or survival. Other sections give a brief synopsis of the elementary mathematical theory of life contingencies, including those involving more than one life; instructions for using the actuarial tables, with numerical examples; and a complete account of the methods and processes used in constructing the life tables. Because of the increasing interest in the preparation of life tables on the part of demographers, public health workers, and other groups, an effort has been made to render this statement of methods and processes intelligible to readers having a reasonable knowledge of mathematics and statistics, but without specific actuarial training. For this reason, some of the explanations will doubtless seem to the actuary unnecessarily full, and even somewhat tedious. An appendix, intended primarily for actuaries, explains the special processes used in the construction of the actuarial tables, and certain other technical matters.

Accuracy of the tables

It is well known that the statistics on which these life tables are based are subject to various errors, the magnitude of which is, in most cases, difficult to estimate with precision. These errors, whether found in statistics of populations, deaths, or births, fall into two general classes: (1) incompleteness or underenumeration, and (2) incorrect reporting of some of the pertinent information, such as age, race, or sex. Very little specific information is available as to the extent

of incompleteness of reporting, except in the case of birth statistics.¹ However, it is believed that the unreported cases constitute, in general, a small percentage of the totals involved, except in the case of data for very young children (including births). In the latter case, a serious attempt has been made to introduce a suitable correction in the process of constructing the life tables.² It should be mentioned also that when death statistics are related to the corresponding population data, as in the computation of rates of mortality, any incompleteness in the enumeration of the population tends to offset whatever deficiency may exist in the reporting of deaths. It is believed, therefore, that errors of incomplete reporting are not likely, in general, to be of sufficient magnitude to seriously affect the life table values for white persons. However, there is some indication that in the rural areas of the South the reporting of Negro deaths may be appreciably less complete than the enumeration of Negroes in the census.³ Since 49 percent of the total Negro population is found in the rural parts of the South, it is possible that mortality rates for Negroes may be somewhat understated. There is a more serious possibility of error in the case of the group of "other races" which includes Indians living on reservations, a class which presents real difficulty from the standpoint of complete reporting and enumeration.

Among the errors due to incorrect reporting, those arising from incorrect statements of age are by far the most important class, as regards the construction of life tables. These errors in age fall into two general types: (1) systematic errors, which arise from a preference for ages ending with certain digits, such as 0, 5, and the even numbers generally, and (2) errors characteristic of particular ages or periods of life. The systematic errors are believed to have been largely eliminated in the graduation of the data described in part V. A typical example of an age error of the second type would be that described by Wolfenden⁴ as "a natural inclination to overstate the age until the attainment of majority, and then to understate at adult ages,

¹ See p. 102.

² See pp. 106-108.

³ U. S. Bureau of the Census, *United States Abridged Life Tables, 1939, Urban and Rural, by Regions, Color, and Sex*, p. 5, June 1943.

⁴ Wolfenden, Hugh H., *Population Statistics and Their Compilation (Actuarial Studies, No. 5)*, p. 27, Actuarial Society of America, New York, 1925

with some overstatement in advanced years." Errors of this sort are not easy to detect, especially if the same type of error occurs in both population and death statistics. Only in one instance, in which the effect was particularly noticeable, has any adjustment been made for such errors in the construction of the life tables in this volume. This point is fully discussed in part V.

Errors in the reporting of race probably are relatively infrequent, except in the case of persons of mixed white and Indian blood. There is no general agreement as to what proportion of Indian blood entitles one to be called an Indian, and it is likely that the information furnished on death certificates may often fail to be consistent in this respect with the definition adopted in the population census. Any error arising from this source could scarcely be of sufficient magnitude to have any appreciable influence on mortality rates for the white population, but could easily have a disturbing effect on those for "other races." It is believed that any errors in the reporting of sex would not be sufficiently numerous to seriously affect any of the life tables.

In addition to errors resulting from actual inaccuracies in the data, there are errors due to chance fluctuation in the number of deaths: that is, what is known as sampling error. This is of importance only in fairly small classes, in which a small variation in the absolute number of deaths in a given age group may make a considerable difference in the rate of mortality. Table A, showing the total enumerated population and the total deaths in the 3-year period in each of the six subdivisions of the population for which separate life tables were prepared, indicates the size of the exposure underlying each life table. Sampling errors tend to be largely corrected by the graduation process, in which the mortality rates in each age group are adjusted so as to bring them into line with those in the neighboring age groups. In any case, it is believed that the effect of sampling error is negligible in the life tables for white persons, and of minor importance in those for Negroes, except at the very old ages. However, it may have significantly affected the results for "other races."⁵

If allowance is made for all the possible sources of error discussed above, the life tables for whites and Negroes are believed to be sufficiently accurate and reliable for all ordinary purposes. However, those for "other races" can be regarded only as reasonable approximations. For reasons explained in part V, this is also true of the life table values for subdivisions of the first year of life in all the tables.⁶

In connection with the accuracy of the tables, it should be clearly understood that the values cannot be considered reliable, in most cases, to anything like the number of decimal places or significant figures shown in the tables. The chief purpose of retaining

TABLE A.—1940 ENUMERATED POPULATIONS, AND TOTAL DEATHS REPORTED IN 1939-1941, BY RACE AND SEX: UNITED STATES

RACE AND SEX	1940 population	1939-1941 deaths
White:		
Male.....	59,448,548	2,048,620
Female.....	58,766,322	1,603,192
Negro:		
Male.....	6,269,038	282,490
Female.....	6,596,480	246,497
Other races:		
Male.....	344,006	13,803
Female.....	244,881	8,211

additional figures beyond those which can be regarded as dependable is to secure a reasonable degree of smoothness in the results. This is always desirable, and in many of the uses to which life tables are put excessive roughness is a serious inconvenience. A further reason exists in the case of the actuarial tables, because of the mathematical relationships which hold between different actuarial functions, such as the values of life annuities and assurances. The actuary wishing to make use of the tables is inconvenienced if, because of excessive rounding, these relationships do not hold with a fair degree of precision.

Comparisons based on the life tables

Variation by race and sex.—The most usual measure of the comparative longevity of different populations is the average duration of life, also called the expectation of life at birth. This is the average number of years lived by the members of a specified cohort, or closed group of persons, assumed to be subject throughout life to the life table rates of mortality. A comparison on this basis is given in table B. This table indicates that females live, on the average, longer than males, white persons longer than Negroes, and Negroes not quite so long as those of "other races." There is, however, some objection to the use of the average duration of life as a standard of comparison because the method of calculating it gives great weight to the relatively large number of deaths occurring in the first year of life. This influence may be entirely eliminated by considering instead the average lifetime remaining to those members of the cohort who survive to age 1. This comparison is presented in table C, which shows, in general, about the same relationships as table B. However, the differences between the corresponding values for Negroes and "other races" are slightly increased now that the effect of the high infant mortality among "other races" is no longer reflected in the figures.

TABLE B.—AVERAGE DURATION OF LIFE IN YEARS, BY RACE AND SEX: UNITED STATES, 1939-1941

RACE	Both sexes	Male	Female
All races.....	63.62	61.60	65.89
White.....	64.02	62.81	67.29
Negro.....	53.85	52.26	55.56
Other races.....	54.35	53.56	55.84

⁵ In connection with the distribution of "other races" deaths by subdivisions of the first year of life, a correction was applied for sampling error. See p. 109.

⁶ See p. 108.

TABLE C.—AVERAGE FUTURE LIFETIME IN YEARS AT AGE 1,
BY RACE AND SEX: UNITED STATES, 1939-1941

RACE	Both sexes	Male	Female
All races.....	65.76	64.00	67.73
White.....	66.84	64.98	68.93
Negro.....	57.15	55.93	58.46
Other races.....	58.90	58.40	60.14

Another possible standard for comparing the longevity of different populations is provided by the median length of life, or "probable lifetime," which is the age at which exactly half the original members of the cohort have died, and half are still alive. In other words, it is the age to which an infant born alive has just an even chance of surviving. The values of the median length of life (shown in table D) are greater in every case than those of the average length of life,⁷ the difference ranging from 3.81 years in the case of Negro females to 8.70 years in the case of females of "other races." The use of the probable lifetime as a measure of longevity results in a somewhat more favorable showing for "other races," as compared with Negroes, than when the average duration of life was used. In fact, the probable lifetime of males of "other races" slightly exceeds that of Negro females. The reverse was true of the corresponding average durations of life.

TABLE D.—MEDIAN LENGTH OF LIFE IN YEARS, BY RACE AND
SEX: UNITED STATES, 1939-1941

RACE	Both sexes	Male	Female
All races.....	69.85	67.68	72.22
White.....	70.86	68.67	73.19
Negro.....	57.86	56.42	59.37
Other races.....	62.67	61.89	64.54

Still another measure of comparative longevity is the number of persons surviving to stated ages in a cohort of, say, 100,000 live births. Such a comparison is presented in table E for survivors to age 21, and in table F for survivors to age 65. These ages have been chosen as representing, respectively, the attainment of manhood or womanhood, and the retirement age prescribed by the Social Security Act. Table E shows that relatively more Negroes reach age 21 than persons of "other races." This reflects higher rates of mortality in the "other races" group over almost the entire age period in question. However, between ages 21 and 65 the relationship is reversed, and the proportion surviving to the latter age is greater among "other races" than among Negroes.

TABLE E.—SURVIVORS TO AGE 21 OUT OF 100,000 LIVE BIRTHS,
BY RACE AND SEX: UNITED STATES, 1939-1941

RACE	Both sexes	Male	Female
All races.....	92,234	91,392	93,116
White.....	92,951	92,098	93,848
Negro.....	87,367	86,494	88,264
Other races.....	82,853	82,412	83,302

⁷ The explanation of this fact and a discussion of the relative merits of different measures of longevity are given on p. 23.

TABLE F.—SURVIVORS TO AGE 65 OUT OF 100,000 LIVE BIRTHS,
BY RACE AND SEX: UNITED STATES, 1939-1941

RACE	Both sexes	Male	Female
All races.....	60,366	55,776	65,523
White.....	63,201	58,305	68,701
Negro.....	37,638	35,371	40,504
Other races.....	46,130	44,689	49,303

In considering the mortality and longevity of the group of "other races," it should be kept in mind that this is a heterogeneous class made up of elements which differ widely both in the general level of mortality and in its incidence by age. The racial composition of the group is shown in table G, and age-specific death rates for the principal races separately appear in table H, together with comparable figures for whites and Negroes.

TABLE G.—POPULATION OF OTHER RACES,¹ BY SPECIFIED RACE
AND SEX: UNITED STATES, 1940

RACE	POPULATION			PERCENT BY RACE		
	Total	Male	Female	Total	Male	Female
Total other races.....	588,887	344,006	244,881	100.0	100.0	100.0
Indian.....	333,969	171,427	162,542	56.7	49.8	66.4
Chinese.....	77,504	57,389	20,115	13.2	16.7	8.2
Japanese.....	126,947	71,967	54,980	21.6	20.9	22.4
Filipino.....	45,563	39,723	5,840	7.7	11.6	2.4
All other.....	4,904	3,500	1,404	0.8	1.0	0.6

¹ All except white and Negro

TABLE H.—DEATH RATES PER 1,000 ENUMERATED POPULATION,
BY AGE, RACE, AND SEX: UNITED STATES, 1939-1941

SEX AND AGE	White	Negro	Indian	Chinese	Japanese	Other
MALE						
0-4.....	13.2	22.8	35.8	13.7	12.1	12.7
5-9.....	1.2	1.6	3.3	1.0	1.1	1.6
10-14.....	1.1	1.7	2.8	1.6	1.3	1.2
15-19.....	1.7	3.7	5.7	3.5	1.8	1.4
20-24.....	2.3	6.4	7.5	4.7	2.6	4.9
25-29.....	2.5	7.8	6.6	5.0	2.9	5.1
30-34.....	3.1	9.7	8.3	6.9	4.8	4.7
35-39.....	4.2	11.4	8.2	9.5	4.5	7.4
40-44.....	6.1	15.7	9.6	12.8	6.0	7.6
45-49.....	9.1	20.8	13.0	17.1	9.4	12.9
50-54.....	13.7	29.4	16.3	23.8	11.4	19.4
55-59.....	20.7	36.1	24.1	38.3	17.6	27.5
60-64.....	30.0	43.8	30.1	47.7	27.4	59.6
65-74.....	53.1	54.5	48.4	80.2	45.7	92.2
75 and over.....	135.0	119.8	109.9	192.1	110.0	103.7
FEMALE						
0-4.....	10.4	18.1	32.1	13.7	9.4	10.7
5-9.....	.9	1.3	2.8	1.9	1.0	3.0
10-14.....	.7	1.5	3.0	1.5	.8	1.4
15-19.....	1.2	4.2	6.4	2.8	1.5	2.7
20-24.....	1.6	5.8	9.5	3.6	1.9	4.5
25-29.....	2.0	6.6	9.1	4.3	3.3	2.5
30-34.....	2.4	8.2	8.4	2.5	2.5	5.5
35-39.....	3.1	9.9	9.4	4.6	3.3	4.4
40-44.....	4.3	14.0	9.6	5.6	3.9	18.5
45-49.....	6.1	17.6	11.2	9.8	6.7	23.2
50-54.....	9.0	25.7	16.0	15.1	7.9	45.3
55-59.....	13.5	32.4	21.9	17.0	13.9	57.5
60-64.....	20.7	40.0	28.0	28.2	17.3	142.9
65-74.....	40.8	44.9	43.0	42.5	37.0	91.7
75 and over.....	120.8	96.5	103.7	93.8	49.0	388.9

¹ Rate based on less than 10 deaths.

A more detailed comparison of life table values by race and sex is offered by figures 1 to 6, in which are plotted graphically the values at all ages of the rate of mortality, the number of survivors out of 100,000 live births, and the average future lifetime for each of the 12 life tables. These graphs bring out certain

FIGURE 1.—ANNUAL RATE OF MORTALITY PER 1,000, FOR EACH RACE BY SEX: UNITED STATES, 1939-1941

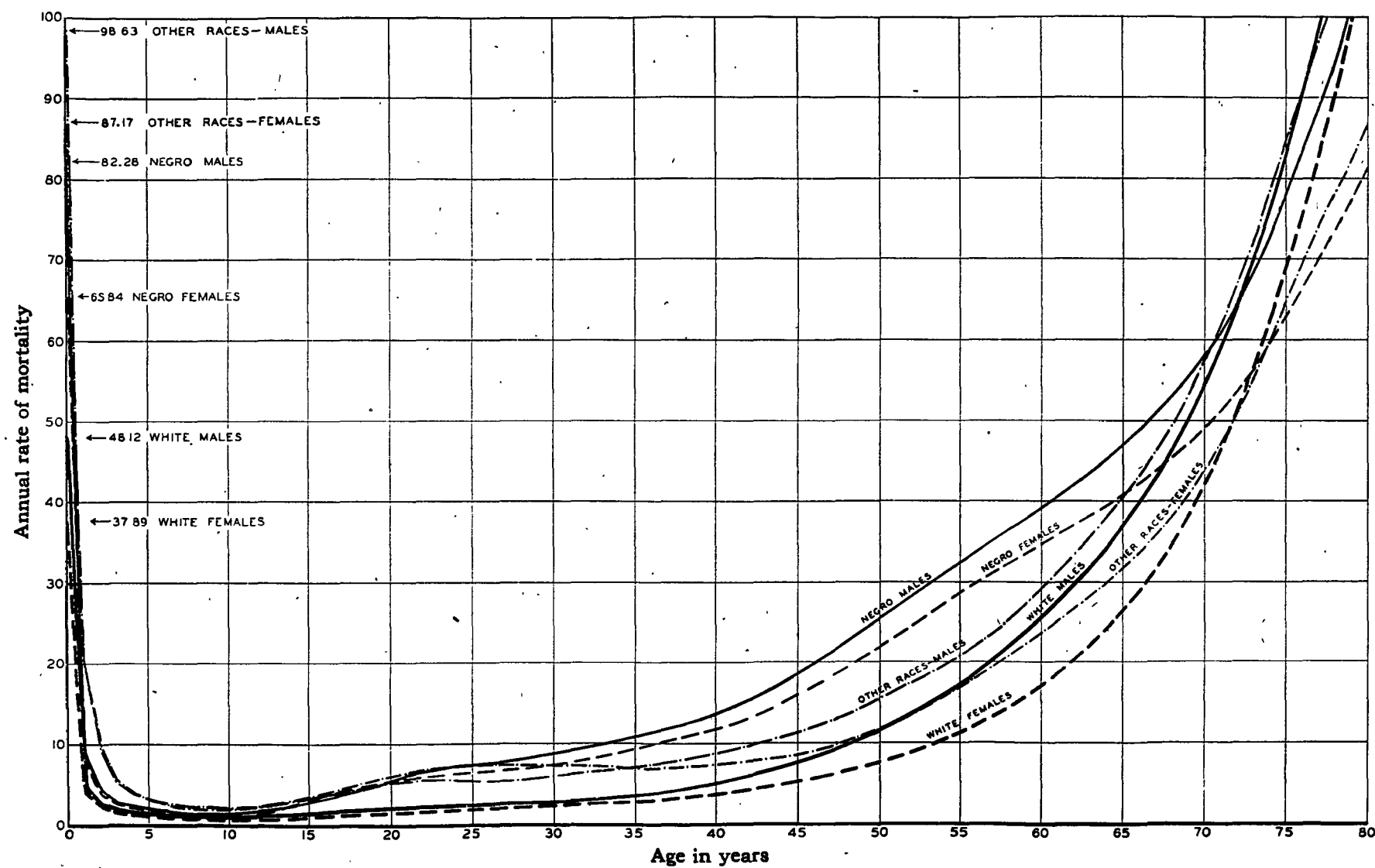


FIGURE 2.—ANNUAL RATE OF MORTALITY PER 1,000, BY RACE AND BY SEX: UNITED STATES, 1939-1941

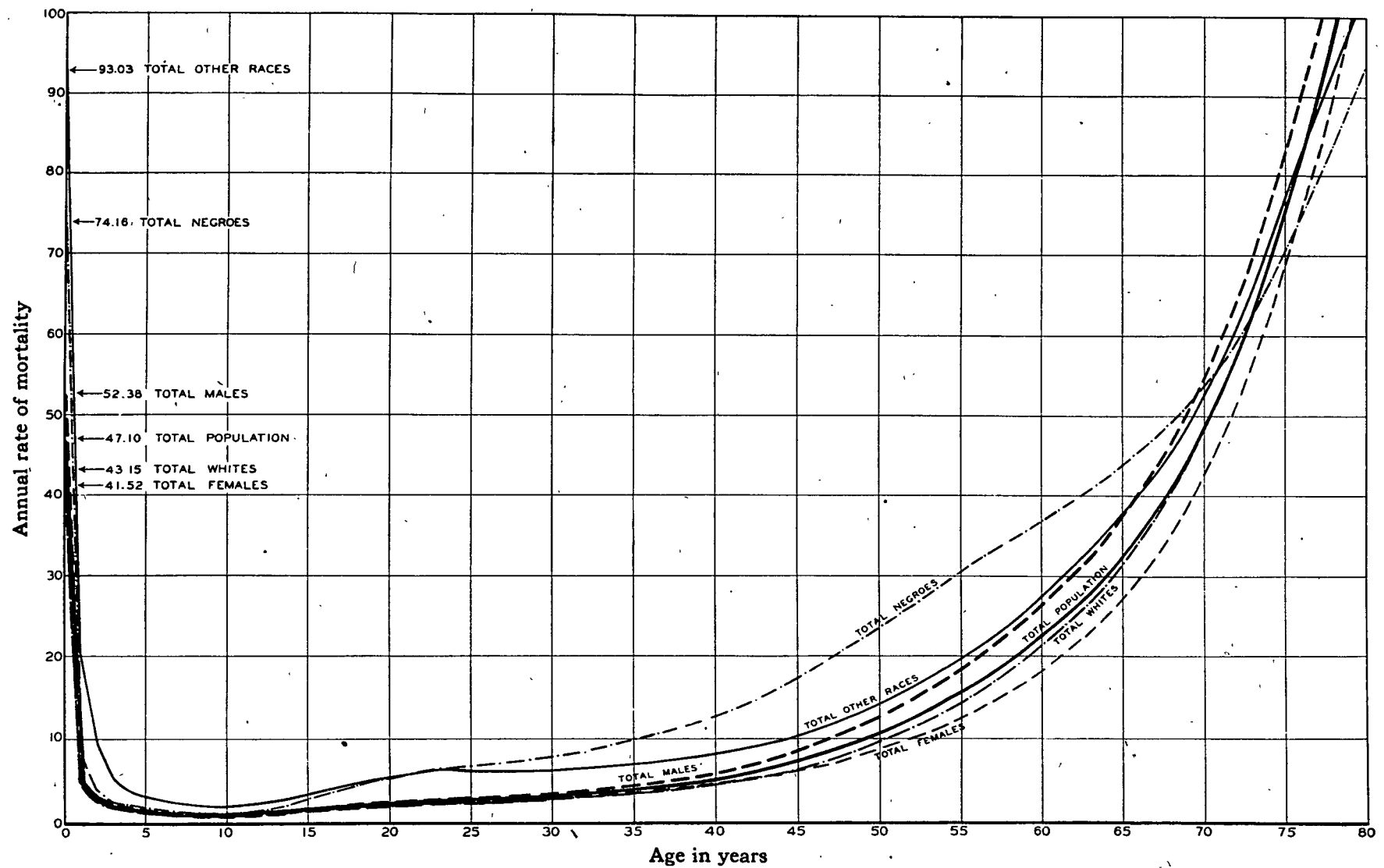


FIGURE 3.—NUMBER OF SURVIVORS OUT OF 100,000 BORN ALIVE, FOR EACH RACE BY SEX: UNITED STATES, 1939-1941

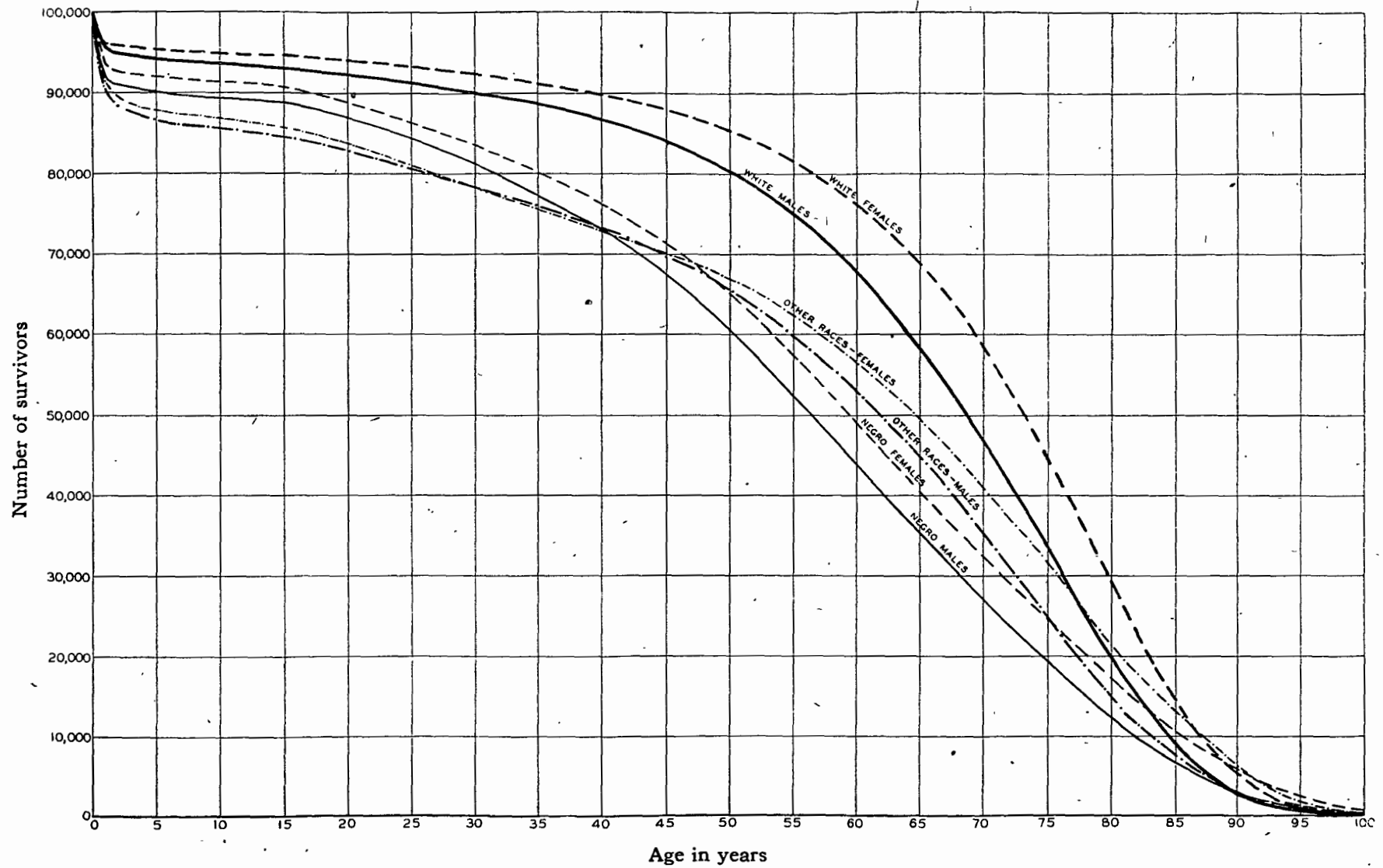


FIGURE 4.—NUMBER OF SURVIVORS OUT OF 100,000 BORN ALIVE, BY RACE AND BY SEX: UNITED STATES, 1939-1941

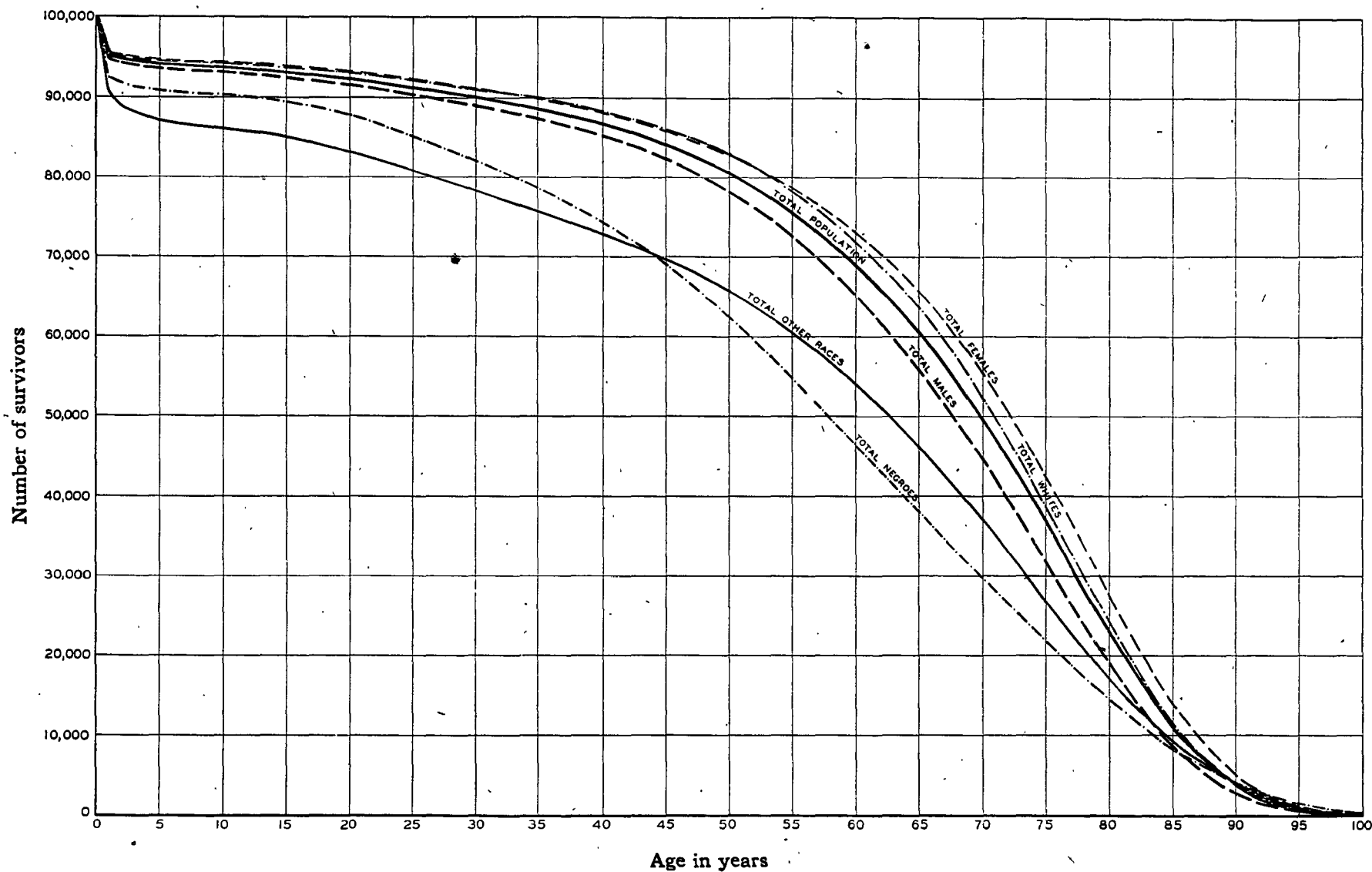


FIGURE 5.—AVERAGE FUTURE LIFETIME, FOR EACH RACE BY SEX: UNITED STATES, 1939-1941

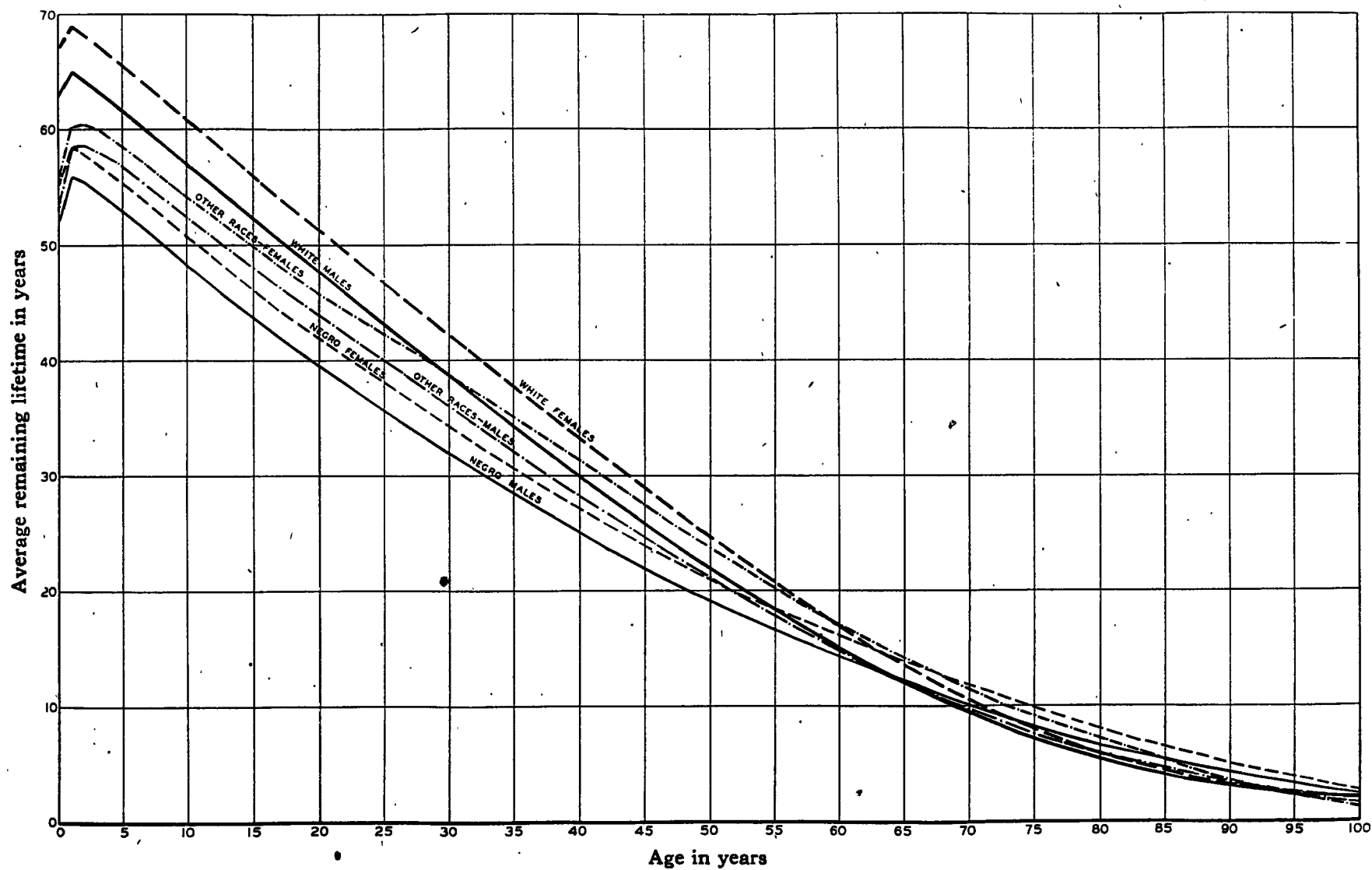
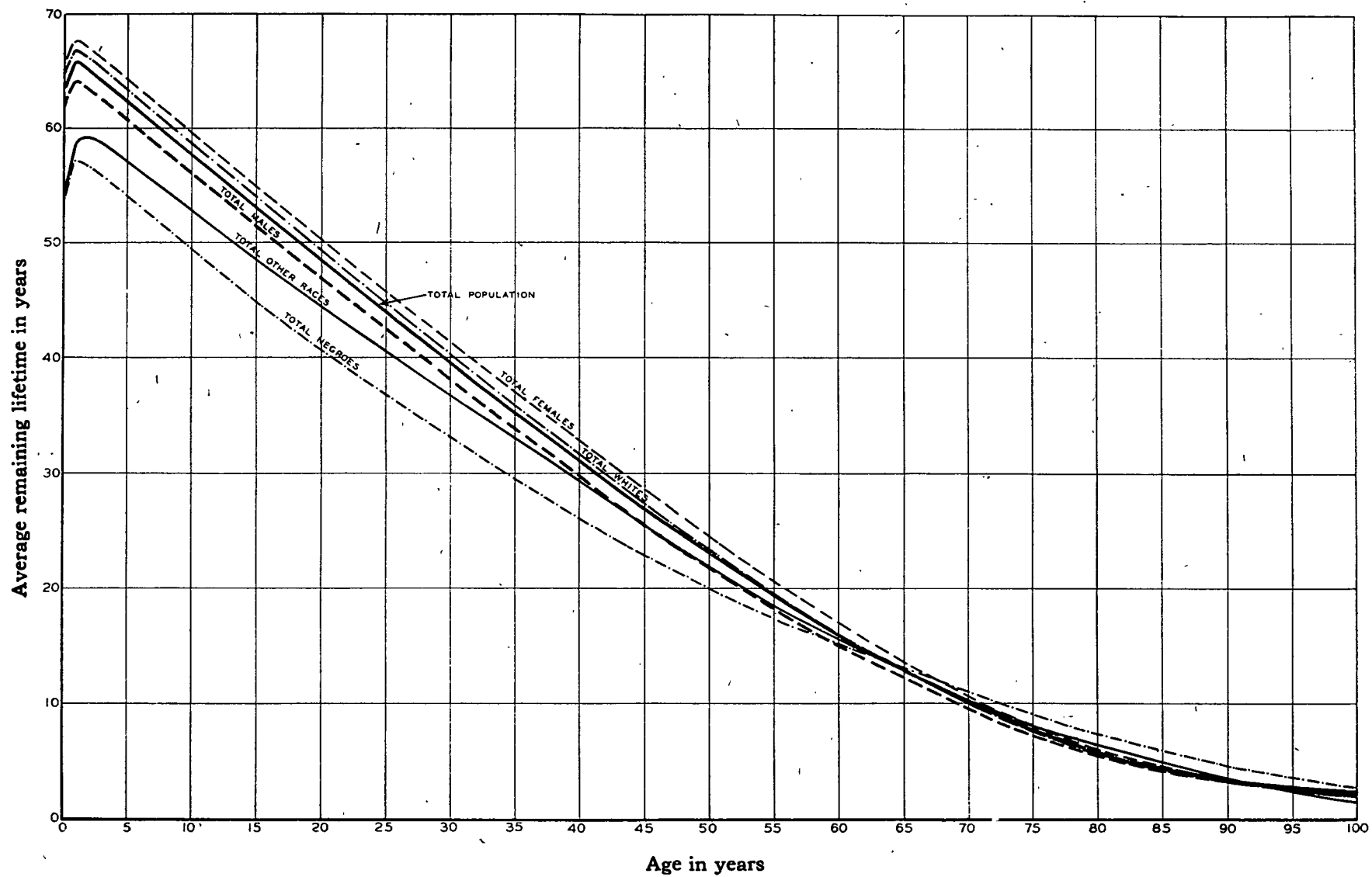


FIGURE 6.—AVERAGE FUTURE LIFETIME, BY RACE AND BY SEX: UNITED STATES, 1939-1941



relationships which may, at first sight, appear somewhat surprising. For example, figure 2 shows that the mortality rates for total females are lower than those for total whites at all ages above 42, notwithstanding the fact that Negro females show a much higher mortality than white males during a large part of this age interval. This seeming inconsistency is due to the fact that Negro and "other races" females form a much smaller group than white males, so that combining them with white females produces less change in the mortality rate of the entire group (as compared with that for white females before the addition) than if the white males had been added.

Because mortality rates for females are so consistently lower, age by age, than the corresponding rates for males, particular interest attaches to the few instances in which exceptions to this general rule occur. The exceptions are found among Negroes at ages 14 to 19, inclusive, and among "other races" at ages 12 to 34. In the case of Negroes, further analysis by particular causes of death shows that the phenomenon is primarily due to the effect of higher mortality from tuberculosis among females at younger ages. Deaths from puerperal causes appear to be a negligible factor at the ages in question. In the case of "other races," examination of data showing a more detailed racial classification makes it clear that the Indians are almost entirely responsible for the higher female mortality, as the Chinese show heavier mortality for males in all age groups and the Japanese show a very slightly higher mortality among females in the late twenties only. In 1940, Indians constituted 66.4 percent of the total female population excluding whites and Negroes, as indicated in table G. Here again, an excess of deaths from tuberculosis among females is the chief factor involved, but in this case deaths from puerperal causes exert a significant, although secondary influence.

The rates of mortality for "other races" show a further peculiarity in that they decrease with increasing age at ages 24 to 26 for males and at ages 29 to 36 for females. Both of the peculiarities mentioned—the higher mortality of females at certain ages, and interruptions in the steady increase of the mortality rate—are usually prominent features of life tables for depressed countries where the general level of mortality is high, such as British India, Japan, and Bulgaria, and they are commonly associated with a high tuberculosis death rate, combined perhaps with a higher mortality from puerperal causes. It is curious to observe, however, that both peculiarities have been consistently noted in the life tables of Canada, a relatively prosperous country having a level of mortality lower, in general, than that of the United States. For example, in the Canadian life table for 1930–1932 the mortality rate of females exceeds that of males at ages 23 to 42, and the mortality rate of males decreases with advancing age at ages 24 to 26. Analysis of deaths by cause indicates that the higher mortality of Canadian females at certain ages

has been primarily due, as in other countries where this occurs, to deaths from tuberculosis. While the over-all death rate for tuberculosis was, in the period under consideration, only slightly higher in Canada than in the United States, the deaths from this cause in Canada are found to be more heavily concentrated among females and at the younger ages.

Comparison with earlier United States life tables.—Table J presents a comparison of values based on the life tables in this volume with those of earlier United States life tables. In addition, rates of mortality for white males and white females are plotted in figures 7 and 8 for five life tables covering the period 1900 to 1941. Although the life tables for periods prior to 1930 do not cover the entire United States, any possible geographic variation in the mortality of white persons could account for only a small part of the spectacular improvement which the comparison shows. In the 40 years between 1900 and 1940 the average duration of life increased by more than 14 years for white males and more than 16 years for white females. The proportion of persons surviving to age 65 has increased by one-half, and the rate of infant mortality has declined to little more than one-third of its value in 1900. Similar improvement is shown throughout childhood and young adulthood and, to a lesser degree, in middle age. The mortality rate at age 40 has diminished to less than half its former value. At older ages the improvement becomes in proportion progressively less, but the recent figures are slightly lower even at the oldest ages shown. The improvement is more marked in the case of females, and remains substantial in amount to a later age than for males.

In the case of Negroes, only the 1929–1931 and 1939–1941 life table values are shown, since the life tables for Negroes show a considerable geographic variation (perhaps due as much to geographic differences in the completeness of registration of Negro deaths as to actual differences in mortality) which makes it inadvisable to present any comparisons not involving identical areas. However, even in the 10-year period between 1930 and 1940, the improvement is striking, the average duration of life of Negroes having risen during the decade nearly 5 years for males and more than 6 years for females.

Figure 7 calls attention to one rather curious feature which calls for special comment. This is the low level of mortality above age 45 in the life table for white males in 1919–1921 in the death-registration States of 1920. From age 52 to 69, the rates of mortality in this life table are actually lower than those of the 1939–1941 table for white males in the United States. The years 1919 to 1921, coming immediately after the influenza epidemic of 1918, were years of unusually low mortality, probably because many persons who, under ordinary circumstances, would have died in these 3 years actually died in 1918. These conditions, of course, affected both sexes and a much broader range

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TABLE J.—LIFE TABLE VALUES FOR SELECTED SPECIFIC AGES, BY SEX: DEATH-REGISTRATION STATES OF 1900 AND 1920, AND THE UNITED STATES, AT 10-YEAR INTERVALS, 1900-1941
[The abbreviation D. R. S. stands for death-registration States]

SEX AND AGE		WHITE									
		Annual rate of mortality per 1,000 (1,000q _x)					Number of survivors out of 100,000 live births (L _x)				
		1939-1941 (U. S.)	1929-1931 (U. S.)	1919-1921 (D. R. S. of 1920)	1909-1911 (D. R. S. of 1900)	1900-1902 (D. R. S. of 1900)	1939-1941 (U. S.)	1929-1931 (U. S.)	1919-1921 (D. R. S. of 1920)	1909-1911 (D. R. S. of 1900)	1900-1902 (D. R. S. of 1900)
MALE		48.12	62.32	80.25	123.26	133.45	100,000	100,000	100,000	100,000	100,000
0.....		4.87	9.93	16.19	28.21	34.47	95,188	93,768	91,975	87,674	86,655
1.....		1.38	2.66	3.95	4.71	6.06	94,150	91,738	88,842	82,972	80,864
5.....		1.00	1.47	2.11	2.38	2.74	93,601	90,810	87,530	81,519	79,109
10.....		1.43	2.13	2.91	2.83	3.34	93,089	90,074	86,546	80,549	78,037
15.....		2.12	3.18	4.27	4.89	5.94	92,293	88,904	84,997	79,116	76,376
20.....		2.43	3.71	5.04	5.54	7.04	91,241	87,371	83,061	77,047	73,907
25.....		2.79	4.13	5.73	6.60	7.99	90,092	85,707	80,888	74,810	71,219
30.....		3.63	5.10	6.69	8.52	9.32	88,713	83,812	78,441	72,108	68,245
35.....		5.13	6.79	7.50	10.22	10.60	86,880	81,457	75,733	68,848	64,954
40.....		7.66	9.29	9.26	12.64	12.63	84,285	78,345	72,696	65,115	61,369
45.....		11.55	12.78	11.74	15.53	15.37	80,521	74,288	69,107	60,741	57,274
50.....		17.37	18.19	16.53	21.50	21.18	75,156	68,981	64,574	55,622	52,491
55.....		25.48	26.44	24.02	30.75	28.59	67,787	61,933	58,498	48,987	46,452
60.....		36.85	38.65	34.99	43.79	41.66	58,305	52,904	50,663	40,802	39,245
65.....		54.54	57.96	54.63	62.14	58.94	46,739	41,880	40,873	31,527	30,640
70.....		83.13	85.26	81.91	92.53	88.43	33,404	29,471	29,205	21,585	21,387
75.....		124.71	129.97	119.73	135.75	133.63	19,860	17,221	17,655	12,160	12,266
80.....		181.04	184.68	182.32	191.11	191.76	9,013	7,572	8,154	5,145	5,252
85.....		248.94	245.50	238.19	255.17	262.78	2,812	2,356	2,568	1,523	1,523
90.....											
FEMALE		37.89	49.63	63.92	102.26	110.61	100,000	100,000	100,000	100,000	100,000
0.....		4.32	8.79	14.59	25.83	31.15	96,211	95,037	93,608	89,774	88,939
1.....		1.10	2.20	3.49	4.47	5.89	95,309	93,216	90,721	85,349	83,426
5.....		.70	1.13	1.79	2.06	2.46	94,890	92,466	89,564	83,979	81,723
10.....		.96	1.64	2.49	2.65	3.39	94,534	91,894	88,712	83,093	80,680
15.....		1.45	2.77	4.33	4.20	5.54	93,984	90,939	87,281	81,750	78,978
20.....		1.82	3.39	5.52	5.22	6.79	93,228	89,524	85,163	79,805	76,588
25.....		2.20	3.74	6.03	6.03	7.72	92,320	87,972	82,740	77,676	73,887
30.....		2.78	4.33	6.42	7.13	8.39	91,211	86,248	80,206	75,200	70,971
35.....		3.68	5.32	6.76	8.03	9.31	89,805	84,256	77,624	72,425	67,935
40.....		5.23	7.02	8.14	9.91	10.63	87,920	81,780	74,871	69,341	64,677
45.....		7.62	9.59	10.67	12.59	13.37	85,267	78,572	71,547	65,629	61,005
50.....		11.28	13.75	14.63	17.93	18.69	81,520	74,321	67,323	61,053	56,509
55.....		17.14	20.63	21.73	25.83	25.06	76,200	68,462	61,704	54,900	50,752
60.....		26.43	31.25	31.68	37.86	36.41	68,701	60,499	54,299	47,086	43,806
65.....		42.33	48.66	50.23	58.63	53.69	58,363	49,932	44,638	37,482	35,206
70.....		68.89	74.60	75.97	82.52	80.39	44,685	37,024	32,777	26,569	25,362
75.....		108.19	117.42	113.41	125.79	121.15	28,882	23,053	20,492	15,929	15,349
80.....		162.94	170.86	170.44	178.32	174.60	14,487	10,937	9,909	7,152	7,149
85.....		231.41	231.51	230.61	247.59	245.32	5,061	3,719	3,372	2,291	2,322
90.....											

SEX AND AGE		WHITE					NEGRO					
		Average future lifetime in years (e _x)					Annual rate of mortality per 1,000 (1,000q _x)		Number of survivors out of 100,000 live births (L _x)		Average future lifetime in years (e _x)	
		1939-1941 (U. S.)	1929-1931 (U. S.)	1919-1921 (D. R. S. of 1920)	1909-1911 (D. R. S. of 1900)	1900-1902 (D. R. S. of 1900)	1939-1941 (U. S.)	1929-1931 (U. S.)	1939-1941 (U. S.)	1929-1931 (U. S.)	1939-1941 (U. S.)	1929-1931 (U. S.)
MALE		62.81	59.12	56.34	50.23	48.23	82.28	87.32	100,000	100,000	52.26	47.55
0.....		64.98	62.04	60.24	50.26	54.61	9.37	16.57	91,772	91,268	55.93	51.08
1.....		61.68	59.38	58.31	55.37	54.43	1.86	2.95	90,082	88,412	52.95	48.69
5.....		57.03	54.96	54.15	51.32	50.69	1.38	2.11	89,393	87,311	48.34	44.27
10.....		52.33	50.39	49.74	46.91	46.25	2.74	4.33	88,610	86,152	43.74	39.83
15.....		47.76	46.02	45.60	42.71	42.19	5.44	8.58	86,968	83,621	39.52	35.95
20.....		43.28	41.78	41.60	38.79	38.52	7.33	10.96	84,227	79,516	35.72	32.67
25.....		38.80	37.54	37.65	34.87	34.88	8.72	12.75	80,979	75,083	32.05	29.45
30.....		34.36	33.33	33.74	31.08	31.29	10.71	14.84	77,221	70,049	28.48	26.39
35.....		30.03	29.32	29.86	27.43	27.74	13.62	18.13	72,780	64,710	25.06	23.36
40.....		25.87	25.28	26.00	23.86	24.21	18.59	22.40	67,346	58,432	21.88	20.59
45.....		21.96	21.51	22.22	20.39	20.76	25.36	27.50	60,495	51,748	19.06	17.92
50.....		18.34	17.97	18.59	17.03	17.42	32.48	33.92	52,426	44,436	16.60	15.46
55.....		15.05	14.72	15.25	13.98	14.35	39.10	41.40	43,533	36,790	14.37	13.15
60.....		12.07	11.77	12.21	11.25	11.51	46.85	50.72	35,371	29,314	12.21	10.87
65.....		9.42	9.20	9.51	8.83	9.03	57.99	70.18	27,236	21,741	10.11	8.78
70.....		7.17	7.02	7.30	6.75	6.84	78.03	92.82	19,456	14,419	8.17	6.99
75.....		5.38	5.26	5.47	5.09	5.10	107.30	129.91	12,186	8,239	6.58	5.42
80.....		4.02	3.99	4.06	3.88	3.81	137.63	177.61	6,444	3,660	5.34	4.30
85.....		3.06	3.03	3.18	2.99	2.85	174.17	220.32	2,836	1,246	4.23	3.42
90.....												
FEMALE		67.29	62.67	58.53	53.62	51.08	65.84	72.04	100,000	100,000	55.56	49.51
0.....		68.93	64.93	61.51	58.69	56.39	7.96	14.37	93,416	92,796	58.46	52.33
1.....		65.57	62.17	59.43	57.67	56.03	1.75	2.84	91,906	90,185	55.40	49.81
5.....		60.85	57.65	55.17	53.57	52.15	1.61	1.61	91,308	89,201	50.75	45.33
10.....		56.07	53.00	50.67	49.12	47.79	3.07	5.12	90,594	88,088	46.13	40.87
15.....		51.38	48.52	46.46	44.88	43.77	5.32	8.82	88,736	85,078	42.04	37.22
20.....		46.78	44.25	42.55	40.88	40.05	6.27	10.34	86,198	81,067	38.20	33.93
25.....		42.21	39.99	38.72	36.96	36.42	7.33	11.59	83,384	76,816	34.40	30.67
30.....		37.70	35.73	34.86	33.09	32.82	9.24	13.22	80,092	72,192	30.71	27.47
35.....		33.25	31.52	30.94	29.26	29.17	11.81	16.25	76,084	67,271	27.19	24.30
40.....		28.90	27.39	26.98	25.45	25.51	16.02	20.18	71,157	61,365	23.89	21.39
45.....		24.72	23.41	23.12	21.74	21.89	21.87	26.65	64,885	54,920	20.95	18.60
50.....		20.73	19.60	19.40	18.18	18.43	28.68	34.99	57,314	47,074	18.38	16.27
55.....		17.00	16.05	15.93	14.92	15.23	34.72	42.20	48,928	38,761	16.10	14.22
60.....		13.56	12.81	12.75	11.97	12.23	40.90	49.35	40,504	30,852	13.93	12.24
65.....		10.50	9.98	9.94	9.38	9.59	49.12	61.74	32,354	23,341	11.82	10.38
70.....		7.92	7.66	7.62	7.20	7.33	62.94	73.41	24,502	16,576	9.81	8.62
75.....		5.88	5.63	5.70	5.35	5.50	81.27	97.84	17,039	10,822	8.02	6.90
80.....		4.34	4.24	4.24	4.06	4.10	105.29	128.34	10,622	6,033	6.41	5.48
85.....		3.24	3.17	3.16	3.00	3.02	141.32	172.03	5,652	2,774	4.96	4.20
90.....												

FIGURE 7.—ANNUAL RATE OF MORTALITY PER 1,000 FOR WHITE MALES: DEATH-REGISTRATION STATES OF 1900 AND 1920, AND THE UNITED STATES, AT 10-YEAR INTERVALS, 1900-1941

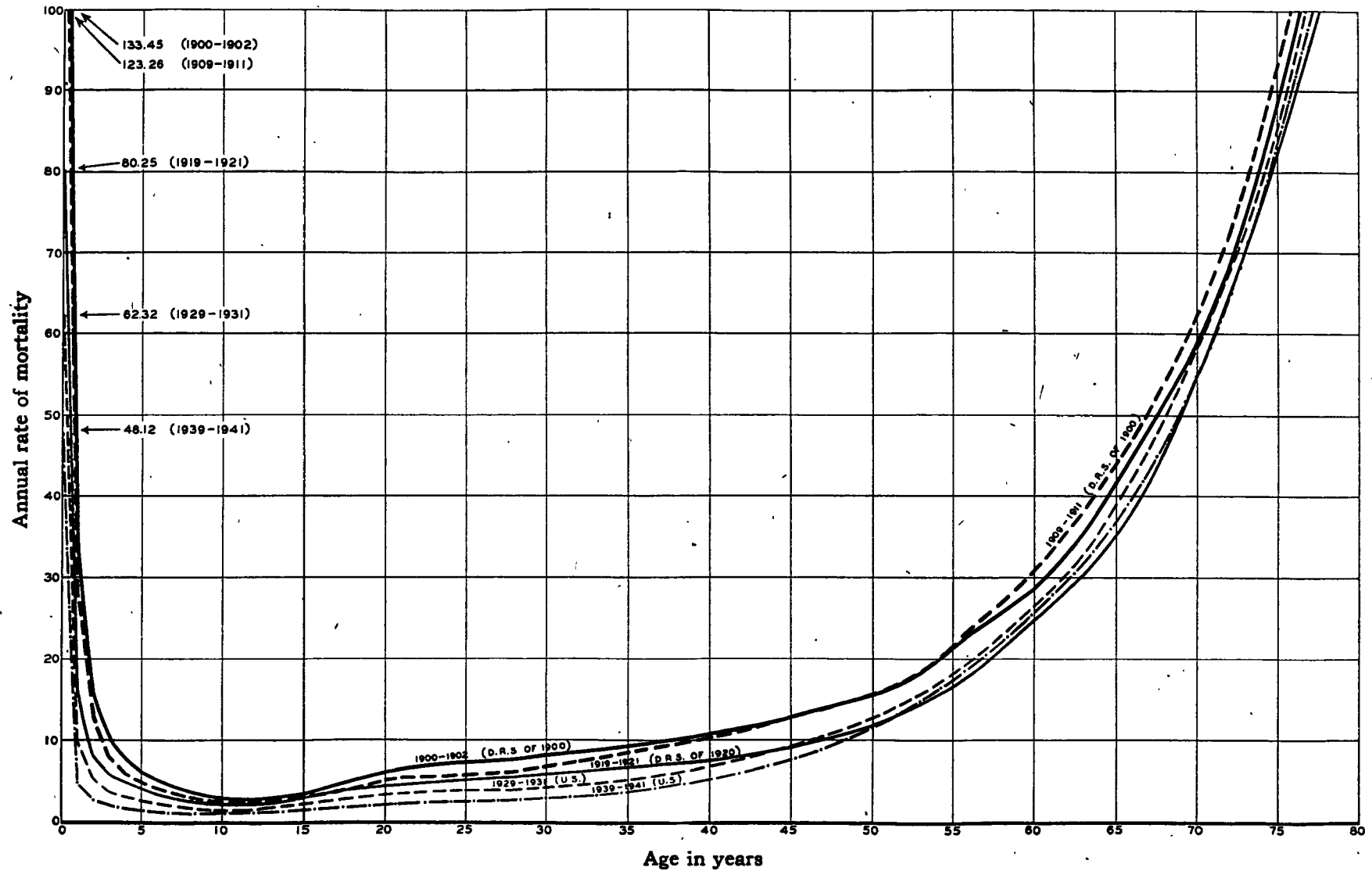
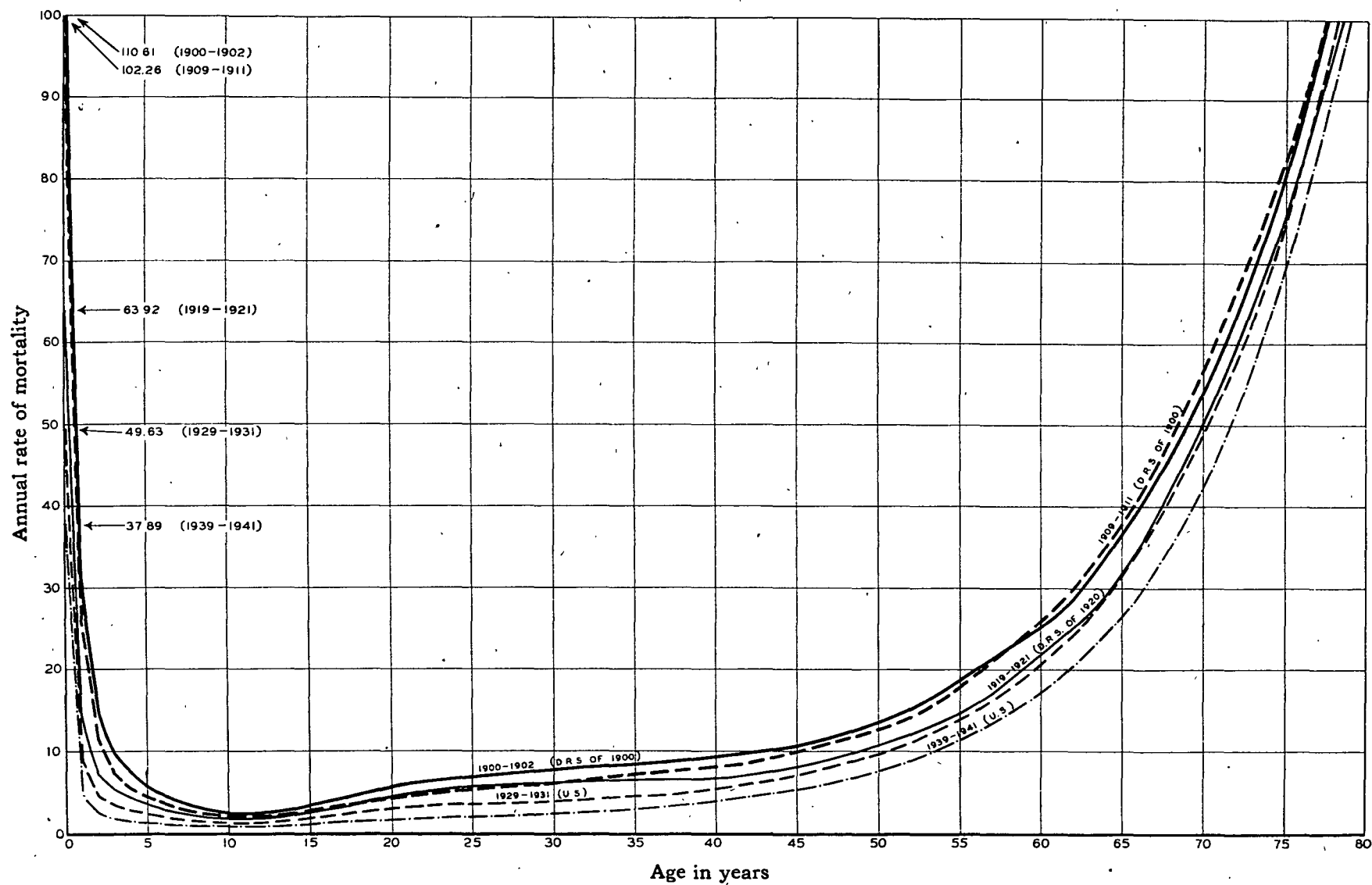


FIGURE 8.—ANNUAL RATE OF MORTALITY PER 1,000 FOR WHITE FEMALES: DEATH-REGISTRATION STATES OF 1900 AND 1920, AND THE UNITED STATES, AT 10-YEAR INTERVALS, 1900-1941



of ages than that indicated above. However, among males at younger ages and among females at most ages, there has been a steady improvement in mortality since 1921, so that the low mark set in the post-epidemic years has since been surpassed. However, in the case of males above age 55 there has been only a slight decline in the mortality rate during the last 20 years, so that the low level of 1919-1921 has not yet been equaled.

Comparison with recent life tables for other countries.—

In tables K, L, and M, life table values for the United States in 1929-1931 and 1939-1941 are compared with the corresponding values for a number of other countries. The selection of these countries has been influenced, to a considerable extent, by the availability of reliable life tables covering recent periods. (Only those for periods ending in or after 1930 have been used.) It is natural, therefore, that the majority of the countries selected are European. Figures 9 and 10 exhibit graphically the number of survivors to successive ages for white males and white females in the United States in 1929-1931 and 1939-1941, in comparison with similar curves for England and Wales, British India, Italy, Japan, Mexico, and New Zealand. These six countries were selected mainly in order to secure a wide range in the general levels of mortality represented. New Zealand and British India, in particular, have long been regarded as representing, respectively, the lowest and the highest general level of mortality among those countries for which reliable life tables are available. Values for the United States death-registration States of 1900 in 1900-1902 have also been plotted, so that the amount of improvement over the 40-year period can be compared with the variation in present conditions as between different parts of the world. These charts show that survival rates in Mexico and British India were still in 1930 far below the level which characterized the United States in 1900, and those of Japan had not quite reached that level. The English life tables of 1930-1932 exhibit lower survival rates up to about age 45 than the United States tables covering approximately the same period. However, at subsequent ages the cumulative effect of the lower adult mortality of England and Wales (which began at age 20) results in a larger number of survivors than in the United States.

Ten years later, in 1939-1941, the United States had not attained the low mortality found in New Zealand about 4 years earlier. This indicates the possibility of still further improvement; and the 1934-1938 life table for New Zealand is by no means to be regarded as reflecting the ultimate low level of mortality which is never to be surpassed.

All the values employed in the preparation of tables K, L, and M and figures 9 and 10 have been obtained from official government life tables except those for Austria and Mexico and some of the values for Canada. The Austrian life table was published by an association of insurance companies, while the Mexican table appeared in a signed article in a journal published by the Department of Public Health. The official Canadian life table commences not at birth but at age 5, so that the numbers of survivors in the life table cohort are not comparable with similar figures for other countries. Accordingly, all these values, as well as the values of the rate of mortality and the average future lifetime at ages 0 and 1, have been taken from an unofficial life table⁸ covering the same period of years. For four countries, it was not possible to secure the original publications, and the figures were obtained from secondary sources. In the case of Denmark and Sweden, the available sources were official yearbooks, while the figures for Austria and Czechoslovakia were obtained from a German Government publication.

The Mexican life table was not graduated, and the rates of mortality were generally overstated at the ages which are multiples of 5 (these being the ages for which values are shown in table K), because of a decided preference for such ages in the reporting of ages at death. For this reason, it would have been unfair to Mexico to use the published mortality rates in the comparison. Accordingly, the values shown in table K have been corrected for this error.⁹

⁸ See list of sources of foreign life table values on p. 20.

⁹ The correction was made by referring to a graduated life table for Mexico, that of J. B. Solórzano as adjusted by Giorgio Mortara (published in *Estadística*, Journal of the Inter American Statistical Institute, vol. 11, No. 5, pp. 78-80, March 1944). For each age x (except ages 0, 1, 5, and 10) for which the mortality rate is shown in table K, a corrected value of the number of deaths at age x in the life table cohort was obtained by accepting as correct the total number of deaths at ages x to $x+4$, and assuming the number of deaths at age x to be the same fraction of the total deaths at ages x to $x+4$ as in the Solórzano-Mortara table. The latter table is for both sexes combined and covers the period 1929-1933.

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TABLE K.—ANNUAL RATE OF MORTALITY PER 1,000, FROM RECENT LIFE TABLES FOR SELECTED COUNTRIES, BY SEX FOR SELECTED SPECIFIC AGES

SEX AND AGE	Australia, 1932-1934	Austria, 1930-1933	Belgium, 1928-1932	British India, 1921-1930	Canada, 1930-1932	Czecho- slovakia, 1929-1932	Denmark, 1931-1935	England and Wales, 1930-1932	France, 1928-1933	Germany, 1932-1934	Italy, 1930-1932
MALE											
0.....	45.43	115.40	100.75	248.7	99.97	148.69	81.47	71.86	90.18	85.35	115.32
1.....	7.75	14.00	17.11	91.8	12.82	19.32	9.01	15.30	16.90	9.26	38.97
5.....	1.84	3.41	3.12	19.3	2.62	3.80	1.34	3.43	2.85	2.32	3.65
10.....	1.19	1.86	1.54	7.9	1.60	1.99	1.13	1.46	1.63	1.33	1.99
15.....	1.49	2.03	2.30	9.8	2.07	2.39	1.47	1.97	2.49	1.57	2.38
20.....	2.19	3.74	4.34	12.7	3.08	4.29	2.56	3.16	5.18	2.83	4.14
25.....	2.49	4.26	3.98	15.3	3.40	4.55	2.66	3.30	5.23	2.97	4.27
30.....	2.71	4.38	4.44	19.3	3.41	4.64	2.68	3.40	5.88	3.24	4.66
35.....	3.46	5.62	5.19	24.1	3.98	5.52	3.24	4.21	7.07	3.94	5.30
40.....	4.60	7.03	6.40	29.4	4.94	7.07	4.01	5.62	8.90	4.82	6.36
45.....	6.59	9.51	8.35	34.9	6.30	9.23	5.84	7.99	11.64	6.52	7.94
50.....	9.66	12.99	11.51	41.0	9.03	12.85	8.32	11.28	15.33	9.39	10.63
55.....	14.93	18.93	16.59	48.1	13.29	17.97	12.44	16.14	20.71	14.18	14.68
60.....	22.16	26.72	24.77	57.9	19.38	25.79	18.66	24.15	29.18	21.72	21.92
65.....	33.11	40.77	37.87	72.7	29.75	38.96	30.97	37.91	42.33	34.04	33.19
70.....	50.82	60.33	58.71	97.6	46.34	59.74	48.25	60.35	64.28	54.01	53.23
75.....	78.08	95.56	91.52	142.7	74.03	93.93	78.37	95.19	101.60	87.40	87.79
80.....	126.59	147.73	142.20	218.0	115.27	143.87	121.81	145.00	152.56	136.68	137.99
85.....	188.64	222.52	218.41	360.8	171.67	212.11	189.55	210.48	234.42	207.69	206.64
90.....	249.86	312.73	327.51	577.0	247.11	289.36	284.52	286.14	303.40	287.73	290.32
FEMALE											
0.....	36.42	92.45	78.55	232.3	83.58	124.57	63.08	54.55	71.62	68.39	102.25
1.....	6.45	13.07	14.78	86.5	13.79	18.57	7.18	13.45	15.13	8.23	39.05
5.....	1.58	3.43	2.68	16.5	2.32	3.76	1.32	2.98	2.79	2.15	3.66
10.....	0.87	1.75	1.50	8.1	1.40	2.10	0.78	1.34	1.60	1.14	1.79
15.....	1.13	1.94	2.40	11.5	1.95	2.50	1.23	1.91	3.04	1.30	2.64
20.....	1.83	3.26	3.79	17.6	2.95	3.85	2.24	2.68	4.82	2.27	3.88
25.....	2.43	3.62	3.81	21.6	3.67	4.37	2.78	2.98	5.00	2.70	4.46
30.....	2.79	3.96	4.06	25.1	3.98	4.48	3.05	3.19	4.78	3.01	4.39
35.....	3.41	4.42	4.48	29.3	4.48	5.02	3.56	3.64	5.14	3.48	4.81
40.....	4.02	5.14	5.22	34.5	5.12	5.69	4.56	4.40	6.08	4.22	5.43
45.....	5.23	7.06	6.49	39.0	6.15	6.89	5.74	5.84	7.50	5.46	6.20
50.....	7.44	9.40	8.69	43.1	8.04	9.50	7.82	8.16	9.77	7.91	8.20
55.....	10.19	13.15	12.42	47.5	11.62	13.49	11.82	11.74	13.38	11.53	11.36
60.....	14.66	19.91	18.81	54.3	17.14	20.51	17.59	17.70	19.26	17.46	17.47
65.....	23.65	32.39	29.69	66.6	26.03	33.08	27.70	27.55	29.86	28.53	28.40
70.....	38.02	51.22	48.12	88.8	40.57	54.30	44.22	44.51	48.13	47.61	46.53
75.....	62.29	85.97	78.97	130.1	67.35	84.42	75.87	74.14	78.75	80.33	79.61
80.....	101.06	131.56	129.67	206.6	107.69	131.28	119.78	118.58	127.93	126.51	127.02
85.....	158.37	202.12	210.38	347.6	160.86	192.59	187.34	179.42	200.02	193.66	191.19
90.....	233.91	279.42	332.04	566.7	228.60	263.29	255.64	250.61	284.63	273.64	267.86

SEX AND AGE	Japan, 1926-1930	Mexico, 1930	New Zea- land, 1934-1938	Scotland, 1930-1932	Sweden, 1931-1935	Switzer- land, 1933-1937	UNION OF SOUTH AFRICA		UNITED STATES		
							Whites, 1935-1937	Nonwhites, 1935-1937	Whites, 1939-1941	Whites, 1929-1931	Negroes, 1939-1941
MALE											
0.....	140.10	223.69	36.53	93.46	54.86	52.42	66.41	183.65	48.12	62.32	82.28
1.....	43.12	105.89	4.71	22.31	7.65	7.77	14.64	70.78	4.87	9.93	9.37
5.....	6.44	16.86	1.72	3.36	1.60	2.15	2.38	7.16	1.38	2.66	1.86
10.....	2.63	6.47	1.00	1.80	1.33	1.23	1.54	3.64	1.00	1.47	1.38
15.....	5.02	5.60	1.47	1.98	1.87	1.73	1.85	5.41	1.43	2.13	2.74
20.....	9.82	10.00	2.18	3.26	3.90	3.26	3.46	8.39	2.12	3.18	5.44
25.....	8.61	10.96	2.34	3.52	3.55	3.47	3.50	9.36	2.43	3.71	7.33
30.....	7.39	12.17	2.32	3.83	3.66	3.43	3.52	10.43	2.79	4.13	8.72
35.....	7.70	13.72	3.03	5.04	4.07	4.33	4.72	12.70	3.63	5.10	10.71
40.....	9.58	16.14	4.41	6.76	4.55	5.50	6.00	15.34	5.13	6.79	13.62
45.....	12.69	18.97	5.84	8.96	6.43	7.86	9.30	17.88	7.66	9.29	18.59
50.....	17.50	22.32	8.43	11.51	8.46	11.67	13.08	21.36	11.55	12.78	25.36
55.....	24.95	27.16	12.56	16.56	11.87	17.47	18.64	25.14	17.37	18.19	32.48
60.....	36.71	36.64	19.51	25.18	17.68	26.11	25.56	35.76	25.48	26.44	39.10
65.....	54.86	46.14	30.17	39.89	27.65	39.48	37.29	46.30	36.85	38.65	46.85
70.....	80.35	71.91	47.89	62.95	44.89	59.35	53.87	60.30	54.54	57.96	57.99
75.....	117.53	91.69	75.60	98.97	70.13	93.10	84.88	79.55	83.13	85.26	78.03
80.....	170.20	138.53	120.01	150.98	116.63	144.80	120.95	114.10	124.71	129.97	107.30
85.....	243.20	185.59	183.86	211.54	186.33	210.90	192.20	175.88	181.04	184.68	137.83
90.....	341.41	218.64	263.84	293.88	286.31	288.00	300.71	276.70	248.94	245.50	174.17
FEMALE											
0.....	124.14	196.75	28.70	73.04	41.82	40.83	53.48	163.00	37.89	49.63	65.84
1.....	42.10	108.70	4.25	20.65	6.74	6.35	14.02	70.62	4.32	8.79	7.96
5.....	7.09	17.12	1.40	3.16	1.54	1.82	2.19	7.71	1.10	2.20	1.75
10.....	3.00	6.46	0.80	1.53	1.18	1.03	1.47	4.07	0.70	1.13	1.04
15.....	7.32	5.24	0.97	2.18	2.04	1.49	1.43	6.90	0.96	1.64	3.07
20.....	10.59	9.24	1.62	2.93	3.37	2.56	2.33	10.50	1.45	2.77	5.32
25.....	9.64	10.12	2.10	3.33	3.47	3.08	2.87	12.13	1.82	3.39	6.27
30.....	8.94	11.34	2.44	3.92	3.34	3.16	3.17	11.98	2.20	3.74	7.33
35.....	9.26	12.40	2.73	4.64	3.87	3.45	4.18	12.59	2.78	4.33	9.24
40.....	10.05	13.06	3.58	5.51	4.14	4.14	4.98	14.80	3.68	5.32	11.81
45.....	10.17	15.28	4.94	7.06	5.60	5.49	6.71	14.12	5.23	7.02	16.02
50.....	12.62	19.72	6.69	9.45	7.41	7.85	9.24	18.62	7.62	9.59	21.87
55.....	16.86	24.27	9.62	13.58	10.23	11.60	12.40	22.35	11.28	13.75	28.58
60.....	24.16	37.49	14.86	20.21	15.48	18.04	17.98	32.16	17.14	20.63	34.72
65.....	37.08	50.20	23.94	30.15	24.30	29.78	28.52	40.54	26.43	31.25	40.90
70.....	57.67	79.05	39.33	48.66	39.14	48.79	42.41	55.37	42.33	48.66	49.12
75.....	88.92	97.00	64.55	79.71	66.92	79.43	67.25	70.51	68.89	74.60	62.94
80.....	138.54	144.72	107.95	125.82	107.88	126.30	108.29	99.20	108.19	117.42	81.27
85.....	213.72	194.25	160.60	190.63	173.66	193.70	167.58	153.49	162.94	170.86	105.29
90.....	322.69	256.61	234.75	265.00	260.74	268.00	248.43	243.86	231.41	231.51	141.32

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE L.—NUMBER OF SURVIVORS OUT OF 100,000 LIVE BIRTHS, FROM RECENT LIFE TABLES FOR SELECTED COUNTRIES, BY SEX FOR SELECTED SPECIFIC AGES

SEX AND AGE	Australia, 1932-1934	Austria, 1930-1933	Belgium, 1928-1932	British India, 1921-1930	Canada, 1930-1932	Czecho- slovakia, 1929-1932	Denmark, 1931-1935	England and Wales, 1930-1932	France, 1928-1933	Germany, 1932-1934	Italy, 1930-1932
MALE											
0.....	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000
1.....	95,457	88,460	89,925	75,126	90,003	85,131	91,853	92,814	90,982	91,465	88,468
5.....	93,887	85,933	87,094	60,161	87,681	81,934	90,322	90,069	88,164	89,654	82,846
10.....	93,193	84,732	86,090	50,467	86,931	80,706	89,758	89,023	87,200	88,793	81,738
15.....	92,609	84,039	85,395	54,112	86,253	79,065	89,280	88,360	86,447	88,244	80,936
20.....	91,797	82,847	84,077	51,203	85,144	78,662	88,423	87,245	84,900	87,298	79,669
25.....	90,711	81,209	82,378	47,787	83,713	76,897	87,272	85,824	82,691	86,032	78,014
30.....	89,566	79,507	80,682	43,931	82,308	75,203	86,119	84,416	80,470	84,715	76,317
35.....	88,248	77,555	78,797	39,461	80,899	73,360	84,910	82,585	77,963	83,234	74,486
40.....	86,539	75,247	76,004	34,563	79,212	71,158	83,472	80,935	74,988	81,481	72,396
45.....	84,276	72,273	73,920	29,439	77,071	68,400	81,613	78,357	71,348	79,285	69,944
50.....	81,061	68,454	70,477	24,348	74,229	64,877	78,999	74,794	66,861	76,322	66,884
55.....	76,504	62,337	65,896	19,476	70,221	60,217	75,191	70,041	61,291	72,147	62,942
60.....	69,950	56,757	59,689	14,933	64,772	54,227	69,804	63,629	54,391	66,293	57,683
65.....	61,292	48,275	51,390	10,773	57,564	46,418	62,177	54,899	45,800	58,106	50,606
70.....	50,066	37,834	40,724	7,036	47,662	36,605	51,610	43,361	35,436	47,059	41,175
75.....	36,588	25,909	28,224	3,848	35,125	25,148	38,135	29,605	23,768	33,479	29,299
80.....	22,223	14,103	15,745	1,514	21,512	13,847	23,064	16,199	12,496	19,122	16,707
85.....	9,752	5,406	6,181	316	9,865	5,444	10,346	6,377	4,527	7,732	6,813
90.....	2,935	1,227	1,374	17	2,833	1,356	2,966	1,609	965	1,966	1,732
FEMALE											
0.....	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000
1.....	96,358	90,755	92,145	76,766	91,642	87,543	93,692	94,545	92,838	93,161	89,775
5.....	94,993	88,321	89,616	62,817	89,017	84,400	92,419	92,024	90,205	91,535	84,107
10.....	94,424	87,148	88,690	59,369	88,220	83,126	91,913	91,082	89,245	90,753	83,019
15.....	93,991	86,452	87,966	56,757	87,590	82,316	91,523	90,420	88,416	90,270	82,227
20.....	93,341	85,375	86,635	52,833	86,564	81,031	90,741	89,383	86,727	89,490	80,908
25.....	92,364	83,918	84,991	47,932	85,154	79,383	89,705	88,133	84,585	88,390	79,223
30.....	91,174	82,381	83,347	42,675	83,542	77,669	88,405	86,792	82,545	87,139	77,478
35.....	89,823	80,738	81,606	37,266	81,840	75,873	87,003	85,353	80,563	85,754	75,754
40.....	88,175	78,841	79,584	31,778	79,919	73,886	85,293	83,690	78,381	84,135	73,860
45.....	86,256	76,620	77,445	26,409	77,756	71,668	83,220	81,660	75,851	82,211	71,777
50.....	83,680	73,654	74,667	21,464	75,127	68,907	80,837	78,958	72,728	79,620	69,332
55.....	80,172	69,796	71,001	17,065	71,685	65,214	76,908	75,290	68,809	76,038	66,164
60.....	75,565	64,471	66,933	13,210	66,840	60,174	71,806	70,204	63,687	70,984	61,803
65.....	69,089	57,053	58,780	9,761	60,304	53,023	64,609	63,046	56,747	63,712	55,510
70.....	59,629	46,620	48,857	6,627	51,382	43,050	54,401	53,144	47,194	53,184	46,455
75.....	46,977	33,399	36,002	3,841	39,697	30,671	40,594	40,040	34,821	39,132	34,323
80.....	31,539	19,416	21,570	1,631	25,748	17,947	24,876	24,869	20,962	23,500	20,517
85.....	16,425	8,114	9,055	367	12,865	7,721	11,283	11,694	9,086	10,323	9,017
90.....	5,808	2,164	2,063	22	4,256	2,225	3,484	3,611	2,430	2,868	2,579

SEX AND AGE	Japan, 1926-1930	Mexico, 1930	New Zea- land, 1934-1938	Scotland, 1930-1932	Sweden, 1931-1935	Switzer- land, 1933-1937	UNION OF SOUTH AFRICA		UNITED STATES		
							Whites, 1935-1937	Nonwhites, 1935-1937	Whites, 1939-1941	Whites, 1929-1931	Negroes, 1939-1941
MALE											
0.....	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000
1.....	85,990	77,631	96,347	90,654	94,514	94,758	93,359	81,635	95,158	93,768	91,772
5.....	78,457	61,485	95,212	87,038	93,055	93,112	90,765	72,210	94,150	91,738	90,082
10.....	76,786	58,131	94,576	85,885	92,324	92,314	89,879	70,378	93,601	90,810	89,393
15.....	75,703	56,471	94,069	85,102	91,660	91,725	89,180	68,942	93,089	90,074	88,610
20.....	72,845	54,413	93,217	84,069	90,477	90,627	88,106	66,702	92,293	88,904	86,968
25.....	69,466	51,660	92,156	82,641	88,840	89,082	86,515	63,764	91,241	87,371	84,227
30.....	66,721	48,774	91,084	81,105	87,278	87,586	85,029	60,723	90,092	85,707	80,979
35.....	64,284	45,724	89,954	79,468	85,718	85,948	83,382	57,387	88,713	83,812	77,221
40.....	61,693	42,472	88,365	77,216	83,536	83,936	81,223	53,549	86,880	81,457	72,780
45.....	58,460	38,895	86,174	74,339	81,802	81,292	78,369	49,309	84,285	78,345	67,346
50.....	54,349	35,031	83,328	70,698	78,956	77,614	74,226	44,759	80,521	74,288	60,495
55.....	49,051	30,940	79,243	66,165	75,179	72,590	68,780	39,881	75,156	68,981	52,426
60.....	42,283	26,573	73,472	59,877	70,044	65,213	61,763	34,471	67,787	61,933	43,833
65.....	33,814	21,590	65,232	51,322	62,975	55,710	53,099	28,086	58,305	52,964	35,371
70.....	24,306	16,615	54,184	40,035	53,076	43,811	42,516	21,564	46,739	41,880	27,236
75.....	14,813	10,879	40,151	26,966	40,346	30,258	30,402	15,205	33,404	29,471	19,456
80.....	7,080	6,352	24,845	14,343	25,536	16,666	18,043	9,396	19,800	17,221	12,186
85.....	2,352	2,660	11,303	5,467	11,946	6,548	8,216	4,518	9,013	7,572	6,444
90.....	454	800	3,395	1,366	3,477	1,647	2,177	1,365	2,812	2,356	2,836
FEMALE											
0.....	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000
1.....	87,580	80,325	97,130	92,696	95,818	95,917	94,652	83,700	96,211	95,037	93,416
5.....	79,866	63,131	96,227	89,249	94,505	94,512	92,210	73,801	95,309	93,216	91,906
10.....	78,053	59,672	95,700	88,217	93,892	93,878	91,355	71,798	94,890	92,406	91,308
15.....	76,523	58,062	95,311	87,500	93,265	93,309	90,722	70,040	94,534	91,894	90,594
20.....	73,069	56,081	94,740	86,380	92,069	92,419	89,939	67,153	93,964	90,939	88,736
25.....	69,366	53,459	93,865	85,054	90,497	91,097	88,778	63,480	93,228	89,524	86,198
30.....	66,215	50,702	92,825	83,561	88,944	89,708	87,456	60,601	92,320	87,972	83,384
35.....	63,287	47,748	91,637	81,814	87,404	88,255	85,934	56,133	91,211	86,248	80,092
40.....	60,312	44,679	90,281	79,802	85,715	86,640	84,005	52,479	89,805	84,256	76,084
45.....	57,345	41,634	88,437	77,418	83,702	84,657	81,680	48,714	87,920	81,780	71,157
50.....	54,285	38,303	85,991	74,384	81,106	82,005	78,605	45,083	85,267	78,572	64,885
55.....	50,534	34,351	82,718	70,418	77,706	78,283	74,577	40,691	81,520	74,321	57,314
60.....	45,819	30,019	78,073	64,951	73,117	72,691	69,342	35,734	76,200	68,462	48,928
65.....	39,593	24,260	71,253	57,553	66,647	65,237	62,127	29,783	68,701	60,499	40,504
70.....	31,544	18,178	61,352	47,782	57,326	54,086	52,314	23,565	58,363	49,932	32,354
75.....	22,099	11,280	47,851	35,058	44,532	39,708	40,224	17,156	44,695	37,024	24,502
80.....	12,5										

INTRODUCTION

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TABLE M.—AVERAGE FUTURE LIFETIME IN YEARS, FROM RECENT LIFE TABLES FOR SELECTED COUNTRIES, BY SEX FOR SELECTED SPECIFIC AGES

SEX AND AGE	Australia, 1932-1934	Austria, 1930-1933	Belgium, 1928-1932	British India, 1921-1930	Canada, 1930-1932	Czecho- slovakia, 1929-1932	Denmark, 1931-1935	England and Wales, 1930-1932	France, 1928-1933	Germany, 1932-1934	Italy, 1930-1932
MALE											
0	63.48	54.5	56.02	26.91	58.46	51.92	62.0	58.74	54.30	59.86	53.76
1	65.49	60.5	61.25	34.68	63.96	59.90	66.5	62.25	58.63	64.43	59.71
5	62.57	58.3	59.21	38.96	62.30	58.19	63.6	60.11	56.47	61.70	59.68
10	58.02	54.1	54.88	36.38	57.96	54.04	59.0	55.79	52.06	57.28	55.46
15	53.36	49.5	50.29	32.85	53.41	49.52	54.3	51.19	47.50	52.62	50.98
20	48.81	45.2	46.04	29.57	49.05	45.29	49.8	46.81	43.30	48.16	46.75
25	44.37	41.0	41.95	26.50	44.83	41.27	45.4	42.54	39.40	43.83	42.69
30	39.90	36.9	37.78	23.60	40.55	37.15	41.0	38.21	35.42	39.47	38.58
35	35.46	32.7	33.61	20.99	36.23	33.02	36.5	33.87	31.47	35.13	34.47
40	31.11	28.7	29.48	18.60	31.98	28.96	32.1	29.62	27.62	30.83	30.39
45	26.87	24.7	25.47	16.40	27.79	25.02	27.8	25.51	23.90	26.61	26.37
50	22.83	21.0	21.61	14.31	23.72	21.24	23.6	21.60	20.33	22.54	22.45
55	19.03	17.4	17.94	12.27	19.88	17.68	19.7	17.89	16.93	18.69	18.70
60	15.57	14.2	14.53	10.25	16.29	14.35	16.0	14.43	13.76	15.11	15.16
65	12.40	11.2	11.43	8.26	12.98	11.32	12.6	11.30	10.86	11.87	11.92
70	9.60	8.6	8.69	6.35	10.06	8.67	9.7	8.62	8.29	9.05	9.05
75	7.19	6.3	6.41	4.61	7.57	6.46	7.2	6.43	6.11	6.68	6.68
80	5.22	4.6	4.65	3.13	5.61	4.73	5.2	4.74	4.44	4.84	4.85
85	3.90	3.3	3.35	1.95	4.10	3.48	3.8	3.50	3.23	3.52	3.52
90	2.99	2.4	2.43	1.12	2.97	2.61	2.8	2.63	2.65	2.63	2.59
FEMALE											
0	67.14	58.5	59.79	26.56	60.23	55.18	63.8	62.88	59.02	62.81	56.00
1	68.67	63.5	63.84	33.48	64.72	61.96	67.1	65.48	62.53	66.41	61.32
5	65.64	61.2	61.63	36.61	63.17	60.21	64.0	63.24	60.32	63.56	61.37
10	61.02	57.0	57.25	33.61	58.72	56.10	59.4	58.87	55.95	59.09	57.15
15	56.29	52.4	52.68	30.04	54.15	51.63	54.6	54.28	51.45	54.39	52.67
20	51.67	48.0	48.43	27.08	49.76	47.40	50.0	49.88	47.40	49.84	48.49
25	47.19	43.8	44.33	24.58	45.54	43.33	45.6	45.55	43.52	45.43	44.47
30	42.77	39.6	40.17	22.30	41.38	39.24	41.2	41.22	39.54	41.05	40.41
35	38.37	35.3	35.97	20.18	37.19	35.10	36.8	36.87	35.45	36.67	36.27
40	34.04	31.1	31.77	18.23	33.02	30.98	32.6	32.55	31.37	32.33	32.14
45	29.74	27.0	27.62	16.43	28.87	26.86	28.3	28.30	27.33	28.02	28.00
50	25.58	22.9	23.55	14.65	24.79	22.83	24.1	24.18	23.39	23.85	23.89
55	21.58	19.1	19.64	12.79	20.84	18.98	20.1	20.23	19.57	19.85	19.91
60	17.74	15.4	15.93	10.81	17.15	15.35	16.4	16.50	15.94	16.07	16.13
65	14.15	12.1	12.57	8.76	13.72	12.06	12.9	13.07	12.57	12.60	12.66
70	10.98	9.2	9.60	6.74	10.63	9.24	9.9	10.02	9.58	9.58	9.61
75	8.23	6.8	7.12	4.86	7.98	6.95	7.3	7.45	7.07	7.09	7.09
80	6.01	5.0	5.20	3.25	5.92	5.12	5.4	5.46	5.09	5.15	5.18
85	4.30	3.6	3.76	2.00	4.38	3.80	4.0	4.00	3.64	3.70	3.78
90	3.05	2.7	2.86	1.18	3.24	2.87	3.0	2.98	2.75	2.72	2.82

SEX AND AGE	Japan, 1926-1930	Mexico, 1930	New Zealand, 1934-1938	Scotland, 1930-1932	Sweden, 1931-1935	Switzer- land, 1933-1937	UNION OF SOUTH AFRICA		UNITED STATES		
							Whites, 1935-1937	Nonwhites, 1935-1937	Whites, 1939-1941	Whites, 1929-1931	Negroes, 1939-1941
MALE											
0	44.82	32.44	65.46	56.0	63.22	60.7	58.95	40.18	62.81	59.12	52.26
1	51.07	40.64	66.92	60.7	65.88	63.0	62.12	48.14	64.98	62.04	55.93
5	51.85	46.97	63.70	59.2	62.89	60.1	59.86	50.27	61.68	59.38	52.95
10	47.93	44.57	59.11	54.9	58.37	55.6	55.43	46.53	57.03	54.96	48.34
15	43.68	40.80	54.42	50.4	53.77	50.9	50.84	42.44	52.33	50.39	43.74
20	40.18	37.25	49.89	46.0	49.44	46.5	46.43	38.78	47.76	46.02	39.52
25	37.01	34.10	45.43	41.7	45.31	42.3	42.24	35.45	43.28	41.78	35.72
30	33.43	30.97	40.94	37.4	41.07	38.0	37.93	32.10	38.80	37.54	32.05
35	29.61	27.88	36.42	33.2	36.78	33.6	33.63	28.81	34.36	33.33	28.48
40	25.74	24.84	32.03	29.1	32.50	29.4	29.45	25.69	30.03	29.22	26.06
45	22.02	21.89	27.78	25.1	28.28	25.2	25.43	22.69	25.87	25.28	21.88
50	18.49	19.05	23.64	21.3	24.21	21.3	21.70	19.74	21.96	21.61	19.06
55	15.21	16.23	19.72	17.5	20.29	17.7	18.21	16.84	18.34	17.97	16.60
60	12.23	13.50	16.06	14.1	16.59	14.3	14.97	14.08	15.05	14.72	14.37
65	9.64	11.05	12.76	11.0	13.15	11.3	11.99	11.71	12.07	11.77	12.21
70	7.43	8.66	9.82	8.4	10.12	8.7	9.34	9.49	9.42	9.20	10.11
75	5.61	6.90	7.36	6.3	7.49	6.5	7.05	7.42	7.17	7.02	8.17
80	4.15	5.17	5.35	4.6	5.37	4.7	5.20	5.50	5.38	5.26	6.58
85	3.02	3.98	3.86	3.5	3.76	3.5	3.67	3.84	4.02	3.99	5.34
90	2.17	3.09	2.79	2.5	2.60	2.6	2.40	2.66	3.06	3.03	4.23
FEMALE											
0	46.54	34.07	68.45	59.5	65.33	64.6	63.06	40.86	67.29	62.67	55.56
1	52.10	41.30	69.46	63.1	67.17	66.4	65.60	47.74	68.93	64.93	58.46
5	53.00	48.19	66.10	61.5	64.09	63.3	63.30	49.99	65.57	62.17	55.40
10	49.18	45.87	61.45	57.2	59.49	58.8	58.87	46.33	60.85	57.65	50.75
15	45.11	42.07	56.69	52.7	54.87	54.1	54.27	42.42	56.07	53.00	46.13
20	42.12	38.46	52.02	48.3	50.55	49.6	49.72	39.13	51.38	48.52	42.04
25	39.23	35.23	47.48	44.0	46.38	45.3	45.33	36.28	46.78	44.25	38.20
30	35.98	32.01	42.98	39.8	42.16	40.9	40.98	33.41	42.21	39.99	34.40
35	32.53	28.84	38.51	35.6	37.85	36.6	36.66	30.36	37.70	35.73	30.71
40	29.01	25.66	34.05	31.4	33.54	32.2	32.44	27.29	33.25	31.62	27.19
45	25.39	22.35	29.70	27.3	29.29	27.9	28.29	24.21	28.90	27.39	23.89
50	21.67	19.09	25.47	23.3	25.14	23.7	24.30	20.96	24.72	23.41	20.95
55	18.09	15.98	21.88	19.4	21.13	19.7	20.47	17.95	20.73	19.60	18.38
60	14.68	12.92	17.49	15.9	17.29	16.0	16.82	15.07	17.00	16.05	16.10
65	11.58	10.38	13.91	12.6	13.71	12.5	13.46	12.58	13.56	12.81	13.93
70	8.88	8.06	10.73	9.6	10.51	9.6	10.50	10.23	10.50	9.98	11.82
75	6.59	6.46	8.02	7.1	7.78	7.1	7.88	8.13	7.92	7.56	9.81
80	4.73	4.76	5.85	5.2	5.62	5.2	5.75	6.08	5.88	5.63	8.02
85	3.30	3.43	4.30	3.8	3.97	3.8	4.13	4.29	4.34	4.24	6.41
90	2.24	2.81	3.00	2.8	2.84	2.9	2.93	2.88	3.24	3.17	4.96

FIGURE 9.—NUMBER OF SURVIVORS OUT OF 100,000 LIVE BIRTHS, FROM RECENT LIFE TABLES FOR SELECTED COUNTRIES

I. MALES

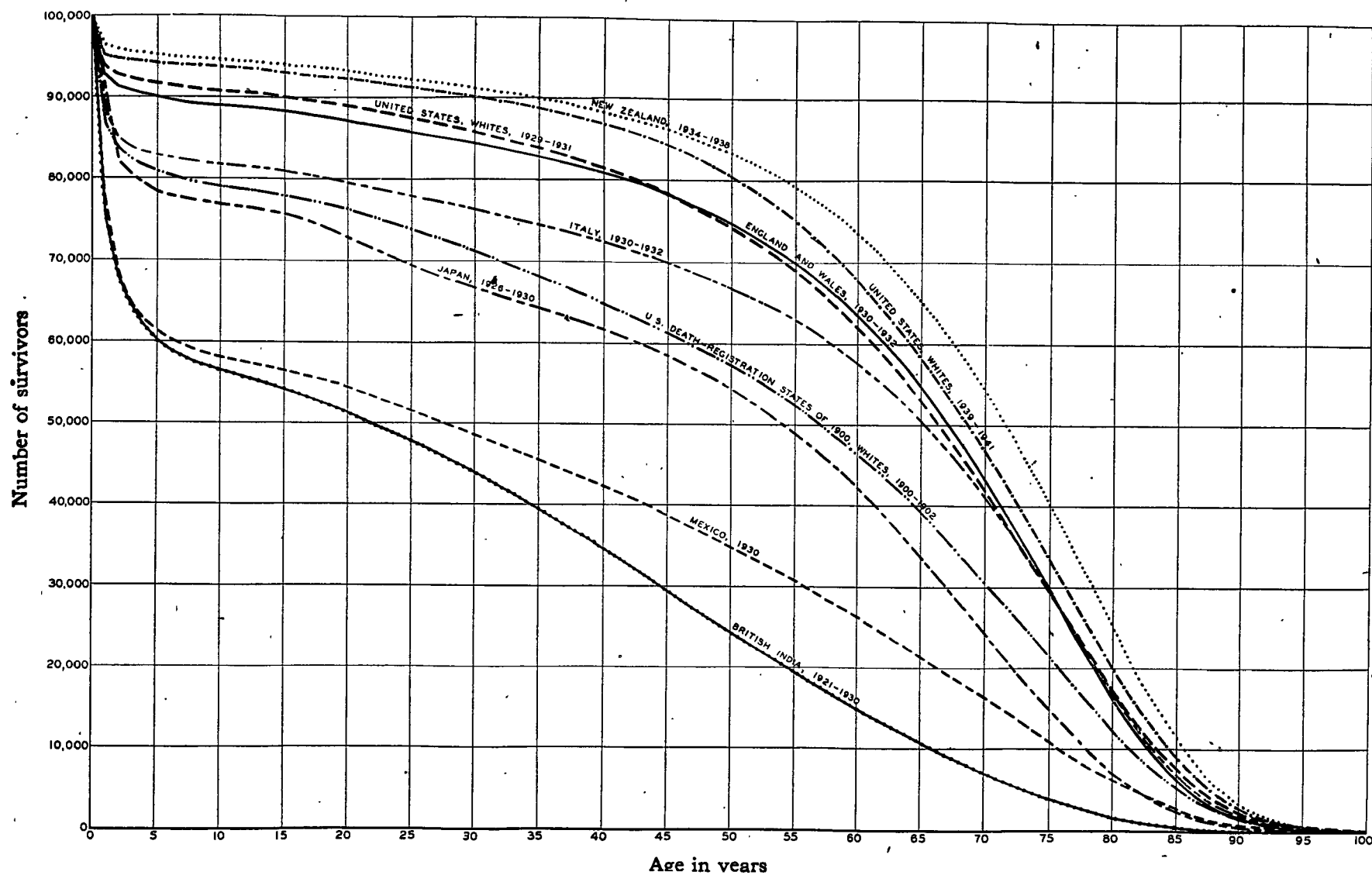
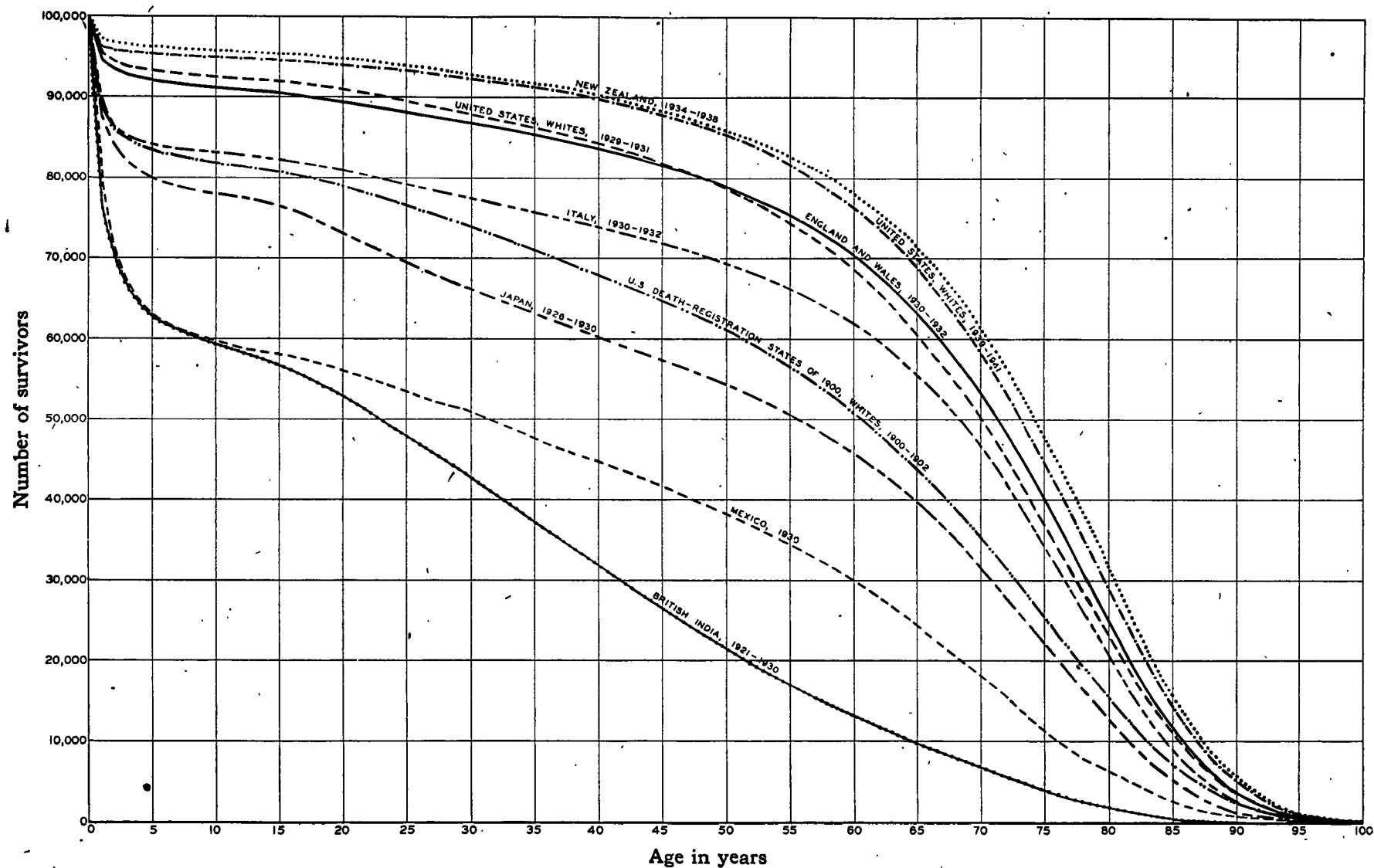


FIGURE 10.—NUMBER OF SURVIVORS OUT OF 100,000 LIVE BIRTHS, FROM RECENT LIFE TABLES FOR SELECTED COUNTRIES

II. FEMALES



INTRODUCTION

SOURCES OF LIFE TABLE VALUES FOR FOREIGN COUNTRIES

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PART II

LIFE TABLES

This part contains the principal life tables presented in this volume. Life tables are given for whites, Negroes, and other races, separately by sex, and for both sexes combined, and also for the total population and for total males and total females. This makes altogether 12 life tables. In addition, table 13 gives, for the same 12 classes and combinations of classes, life table values for certain subdivisions of the first year of life. All these tables are based on the 1940 census of population and the deaths of the 3-year period 1939-1941.

Explanation of the columns of the life table

Both the descriptive titles and the conventional actuarial symbols appear at the head of the columns in each of the tables. The description which follows gives a more detailed explanation of each column of the life table, and may be helpful to some readers.

*Column 1—Year of age (x to $x+1$).—*The year of age, shown in column 1, is the interval between two successive birthdays. For instance, "4-5" indicates the interval between the fourth birthday and the fifth, in other words, the fifth year of life.

*Column 2—Mortality rate ($1,000 q_x$).—*This column shows the number of deaths within 1 year after the birthday indicated, among 1,000 persons alive on that birthday. For example, the rate of mortality at age 45 for white males (table 5) is 7.66 per 1,000. In other words, during 1939-1941, 7.66 out of every 1,000 white males who were alive on their forty-fifth birthday died before reaching age 46. The rates of mortality form the basis of the life table, all the other columns being derived from them.

*Column 3—Number living (l_x).—*This column shows the number of persons who would survive to each age out of a cohort of 100,000 live births, subject throughout life to the rates of mortality shown in column 2. Thus, table 5 shows that out of 100,000 white male babies born alive, 95,188 will complete the first year of life and enter the second; 94,724 will begin the third year; 92,098 will reach age 21; and 33,404 will live to age 75.

*Column 4—Number dying (d_x).—*This column shows the number dying in each successive year of age out of 100,000 live births. Out of 100,000 white males born alive (table 5), 4,812 die in the first year of life, 464 in the second year, 195 in the twenty-first year, and 2,762 in the seventy-fifth year. Each figure in column 4 is the difference between two successive figures in column 3.

*Columns 5 and 6—Stationary population (L_x and T_x).—*Suppose that a group of 100,000 individuals like that assumed in columns 3 and 4 is born every year, each such group being subject throughout life to the rates of mortality shown in column 2. If there were no migration and if the births were evenly distributed over the calendar year, the survivors of these births would make up what is called a stationary population because in such a population the number of persons living in any given age group would never change. When an individual left the group, either by death or by growing older and entering the next higher age group, his place would immediately be taken by someone entering from the next lower age group. Thus, a census taken at any time in such a stationary community would always show the same total population and the same numerical distribution of that population among the various ages. In such a stationary population, column 3 shows the number of persons who, each year, reach the birthday indicated in column 1 while column 4 shows the number who die each year in the indicated age interval.

Column 5 shows the number of persons in the stationary population in the indicated age interval. For example, the figure given for white males in the year of life 45-46 is 83,962. This means that in a stationary population of white males supported by 100,000 annual births and subject always to the rates of mortality shown in column 2, a census taken on any date would show 83,962 persons between 45 and 46 years old.

Column 6 shows the total number of persons in the stationary population (column 5) in the indicated age interval and all subsequent age intervals. For example, in the stationary population of white males referred to in the last illustration, column 6 shows that there would be at any given moment a total of 2,180,567 persons who have passed their forty-fifth birthday. The population at all ages 0 and above (in other words, the total population of the stationary community) would be 6,281,188.

*Column 7—Average future lifetime (e_x).—*The average future lifetime (also called the complete expectation of life) at any age is the average number of years remaining to be lived by those surviving to that age, on the basis of a given set of mortality rates. The values in column 5 can also be interpreted in terms of a single life table cohort, without introducing the concept of the stationary population. From this point of view, each figure in column 5 represents the total time (in years)

lived between the indicated birthdays by those reaching the earlier birthday among the survivors of a cohort of 100,000 live births. Thus, the figure 83,962 for white males in the year of life 45-46 is the total number of years that will be lived between the forty-fifth and forty-sixth birthdays by the 84,285 (column 3) who reach their forty-fifth birthday out of 100,000 white males born alive. The corresponding figure in column 6 (2,180,567) is the total number of years that will be lived after attaining age 45 by the 84,285 reaching that age. This number of years divided by the number of persons (2,180,567 divided by 84,285) gives 25.87 as the average future lifetime of white males at age 45.

Care must be exercised in drawing conclusions from the figures in column 7. Thus, observing that the "expectation of life" at birth is always greater for white persons than for Negroes, one should not conclude that the oldest ages reached by white persons necessarily exceed those attained by the most long-lived Negroes. The difference in the average length of life is due to the fact that a greater proportion of Negroes die before reaching old age. For example, the number surviving to age 65 out of 100,000 born alive is far greater among whites than among Negroes; yet the average length of life remaining at age 65 is practically the same for both races.

Table 13—Subdivisions of the first year of life.—What has been said about the various columns of the life table applies also, with certain obvious modifications, to the life table values for subdivisions of the first year of life, given in table 13. The figures corresponding to age "2-3 weeks" for white males may be taken as an illustration. The age interval (column 1) is the period beginning with the exact age 2 weeks and extending up to the exact age 3 weeks: in other words, the third week of life. The mortality rate of 1.64 in column 2 means that out of every 1,000 white male infants alive exactly 2 weeks after birth during 1939-1941 this number, on the average, died during the following week. The number living (97,194 in column 3) signifies that this many would still be alive exactly 2 weeks after birth out of the life table cohort of 100,000 live births, on the assumption that the mortality rates shown in column 2 have prevailed during the first 2 weeks of life. The number dying (159 in column 4) means that out of the 97,194 alive exactly 2 weeks after birth this number would die during the following week. The figure 1,861 in column 5 indicates that during the third week of life the survivors of the life table cohort of 100,000 white male births have lived a total of 1,861 person-years of life. Or, alternatively, this figure is the number of infants aged 2-3 weeks in a stationary population of white males supported by 100,000 annual births and subject always to the mortality rates shown for white males in column 2 of this table and of table 5. The figure 6,277,446 in column 6 represents the total number of person-years of life lived beyond the first 2 weeks of life

by all the 97,194 survivors to the age of exactly 2 weeks in the life table cohort which started with 100,000 white male births. Alternatively, it is the entire population at all ages beyond 2 weeks in the stationary population already referred to. Finally, the average future lifetime of 64.59 shown in column 7 is the average number of years lived beyond the first 2 weeks of life by the 97,194 survivors to the age of exactly 2 weeks in the life table cohort.

Use of life tables in estimating and forecasting populations

One of the most important applications of life tables in demographic research is their use in estimating the age distribution of a population on a given postcensal date. In particular cases, this may be either a past, present, or future date. While an exhaustive discussion of the subject would be beyond the scope of this volume,¹ an outline of the general procedure will be given. Basically this consists, in the usual method of population projection, in multiplying the number enumerated at each age in the census by a survival rate derived from a life table, in order to obtain the estimated number of survivors on the given date. It is usually most appropriate to obtain the survival rates from the L_x column of the life table (column 5 of the tables on pp. 26 to 49). For example, suppose that in a certain group of white males there were enumerated, in the 1940 census, 32,000 at age 47 on the last birthday, and that it is desired to estimate the number of survivors just 6 years later (that is, on April 1, 1946), on the supposition that the mortality during the 6-year period will be approximately the same as that indicated at the ages in question by the 1939-1941 life table for white males. Now the original group of 32,000 presumably included persons at all ages between exact age 47 and exact age 48, and was, therefore, similar in its age composition to the group at age 47 on the last birthday in the stationary life table population, which numbered 82,568. Now, since the hypothetical life table population does not change with the passage of time either in its total number or in its age composition, the survivors 6 years later of this group of 82,568 would be merely the number at age 53 in the life table population, which is 76,953. Therefore, the survival rate to be applied to the group of 32,000 is 76,953 divided by 82,568, which is .93200; and the estimated number of survivors is 32,000 multiplied by .93200, which gives 29,824. In algebraic terms, the L_{x+6} persons aged $x+6$ in 1946 are the survivors of the L_x persons aged x in 1940. Therefore, the survival rate to be applied to the population at age x is L_{x+6}/L_x .

If migration during the 6 years is thought to have been a significant factor, it is of course necessary to

¹ For a detailed discussion of the subject, see *Estimates of Future Population of the United States, 1940-2000* (prepared by Warren S. Thompson and P. K. Whelpton, and issued by the National Resources Planning Board), Government Printing Office, Washington, D. C., 1943.

obtain some information or to make some assumption as to the number and age composition of the net migrants each year, and to adjust the number of survivors accordingly.

Estimation of the populations at ages under 6 on April 1, 1946, would require a knowledge of the number of births during each of the 6 years. For example, suppose 51,000 white males entered the group through birth during the year April 1, 1943, to April 1, 1944. On April 1, 1946, the survivors of these births would be between exact ages 2 and 3. Now, in the life table population, the number of births during any year is the radix ² of the life table—in this case, 100,000—while the number of survivors on a date just 2 years after the end of the year in which the births occurred would be merely the number at age 2 in the life table population (or 94,592 in column 5). Therefore, the survival rate to be applied to the 51,000 births is 94,592 divided by 100,000 which is .94592; and the estimated number of survivors is 51,000 multiplied by .94592, which gives 48,242. In algebraic terms, the survival rate to be applied to the births of the n th year preceding the date of the estimate is L_{n-1}/l_0 .

In the original example of the 32,000 enumerated at age 47, suppose it had been desired to estimate the number of survivors 6 months later, on October 1, 1946. These individuals would then be at ages ranging from exact age 53½ to exact age 54½. Now the number of persons between these ages in the life table population is approximately l_{54} (column 3): that is, the number of survivors to age 54 out of the life table cohort of 100,000 live births, as indicated in column 3. In this particular case, the figure is 76,380. Therefore, the survival rate to be applied is l_{54}/L_{47} , or 76,380 divided by 82,568, which is .92506; and the estimated number of survivors is 32,000 multiplied by .92506, which gives 29,602.

If the population data are given in 5-year age groups, or can be combined into such groups, it is possible to shorten the arithmetic with very little loss of accuracy by using an average survival rate for each 5-year age group as a whole. Thus, the survival rate over a 6-year period for the age group x to $x+4$ would be $(T_{x+5} - T_{x+10}) \div (T_x - T_{x+5})$. Other situations which may arise can be dealt with along similar lines.

The life table as a frequency distribution

The ages at death in the hypothetical life table cohort (as shown in column 4 of the life tables on pp. 26 to 49) constitute a frequency distribution. In the following discussion, the case of the life table for white males (table 5) will be taken as an illustration, but the remarks to be made will apply equally to all the life tables, except for some difference in the ages and numerical values quoted. The frequency distribution based on the white males life table is exhibited graphically in

figure 11. Perhaps the most obvious characteristic of this distribution is that it is bimodal: that is, it has two modes or maxima, one in the year of age 0-1 and another in the year of age 75-76. The mode at age 0-1 is the higher, more deaths occurring in this than in any other single year of age. It is also clear that the frequency distribution is decidedly skewed toward the left: that is, the frequencies rise very gradually from the "trough" at age 10 to the "peak" at age 75, and then drop off sharply above age 75. The arithmetic mean of the distribution is the average age at death in the hypothetical cohort, or in other words, the average duration of life. Its value in this case is 62.81 years (column 7 of the life table). It is clear that the value of the arithmetic mean is very much influenced by the large number of deaths in the first year of life. If the deaths occurring in the first year were excluded from the distribution, the average age at death of the remaining 95,188 individuals would be one plus the average future lifetime at age 1: that is, 65.98 years. This represents a difference of more than 3 years in the value.

The median of the distribution (that is, the value which has the same number of elements on either side of it) is the median length of life, or probable lifetime, another possible measure of longevity to which reference was made in part I.³ Since the distribution of ages at death in a life table cohort is always characterized by a greater dispersion below the median value than above it, the median always exceeds the arithmetic mean. In the particular case under consideration, the median is 68.67 years, which exceeds the mean value by 2.69 years.

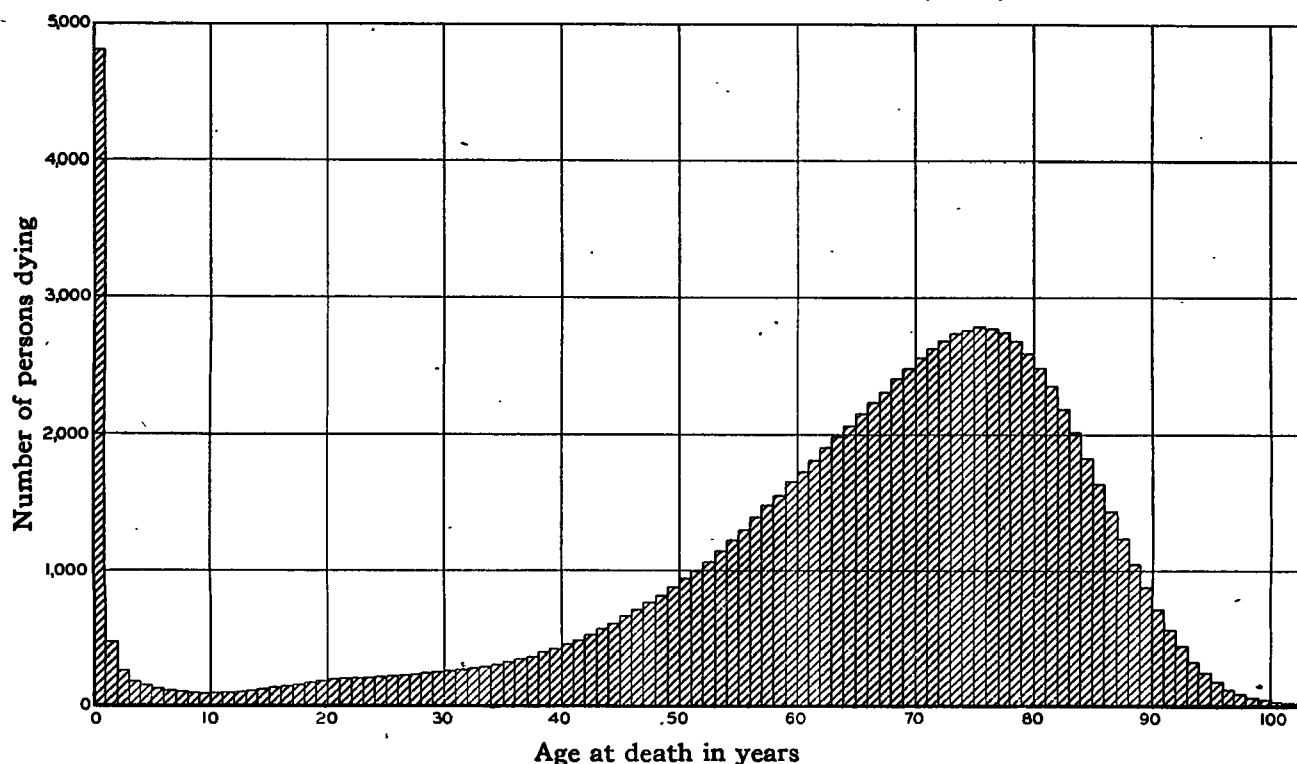
In part I the longevity of different subdivisions of the population was compared also by means of a third criterion, the number of persons surviving to specified ages in the hypothetical cohort. Which of these is the better measure of longevity is a question that cannot be answered categorically. The answer perhaps depends primarily on the purpose such a measure is intended to serve. Certainly no one figure can contain within itself all the information which is provided by the complete frequency distribution.

In view of the pronounced skewness of the distribution, it may be felt that the arithmetic average is not sufficiently representative. The layman, in inquiring what is the "life expectancy" of a newborn infant, probably has in the back of his mind the idea of an age to which the infant has a reasonably good chance of surviving. If he is told that the infant's "expectation of life" is 62.81 years, he may be surprised to be told later that more than 62 percent of white male infants alive at birth outlive their expectation of life while less than 38 percent die before reaching that age. The alternative statement that 68.67 years is the probable lifetime, the age to which the infant has a fifty-fifty chance of surviving, is probably a more satisfactory answer to the layman's question.

² The radix of a life table is the number of births with which the life table cohort begins, or, in algebraic terms, the value of l_0 . In the tables on pp. 26 to 49, this is shown in column 3 opposite the year of age 0-1, and is always 100,000.

³ See p. 3.

FIGURE 11.—FREQUENCY DISTRIBUTION OF AGES AT DEATH IN A COHORT STARTING WITH 100,000 LIVE BIRTHS, BASED ON THE MORTALITY OF WHITE MALES: UNITED STATES, 1939-1941



On the other hand, the objection may be made that the probable lifetime is not sufficiently sensitive to changes in the ages at death of the members of the life table cohort. In fact, its value is not affected by any change in which the age at death of an individual is not actually shifted from one side to the other of the probable lifetime itself. If, for example, the deaths of the 4,812 dying before age 1 in the white males life table were equally spread over all the years of age between birth and age 68, many of these individuals would live much longer; yet the value of the probable lifetime would be unchanged. However, the effect of transferring deaths from one age to another in the hypothetical life table cohort is not entirely relevant, since the mortality rates in the life table were not obtained by observing a single cohort over a period of time, but rather by observing many cohorts simultaneously, a different one at each age. Therefore, the important thing is the effect of a specified change in the rate of mortality at a particular age, without reference to any offsetting change elsewhere. Any change in the mortality rate at any age less than the probable lifetime (unaccompanied by other changes) will alter the value of the probable lifetime. However, changes in mortality rates at ages greater than the probable lifetime will have no effect whatever on its value. Similar remarks apply to the third criterion suggested, the number of survivors to a designated age. The value of the average duration of life, on the contrary, is affected in some measure by any change in the rate of

mortality at any age, or in the ages at death in the life table cohort.

Use of the life table in studying there productive capacity of populations

Another important application of life tables in demographic research is their use in conjunction with fertility rates in investigating the inherent capacity of a population to reproduce itself. This is studied, for the most part, by means of certain specific measures devised for that purpose, the most important of which are the gross and net reproduction rates⁴ and the true rate of natural increase.⁵ While life table survival rates are an important component in the calculation of these measures, they involve other considerations of a highly technical nature, which are outside the scope of this volume.

Mathematical notation employed

One of the mathematical symbols used in the headings of table 13 represents a departure from the standard notation in use by actuaries. This is the symbol

⁴ See Robert R. Kuczynski, *The Balance of Births and Deaths*, 2 vols., The Macmillan Co., New York, 1928; *Fertility and Reproduction*, Falcon Press, New York, 1932; *The Measurement of Population Growth*, Oxford University Press, New York, 1936; D. V. Glass, *Population Policies and Movements in Europe* (Appendix), Oxford University Press, London, 1940.

⁵ See Louis I. Dublin and Alfred J. Lotka, *Length of Life*, The Ronald Press Co., New York, 1936; *On the True Rate of Natural Increase*, *Journal of the American Statistical Association*, vol. 20, No. 151, pp. 305-339, September 1925; Alfred J. Lotka, *The Geographic Distribution of Intrinsic Natural Increase in the United States, and an Examination of the Relation Between Several Measures of Net Reproduction*, *ibid.*, vol. 31, No. 194, pp. 273-294, June 1936; *Some Recent Results in Population Analysis*, *ibid.*, vol. 33, No. 201, pp. 164-178, March 1938. See also Glass' book cited in the preceding footnote.

q_x , which appears in the heading of column 2 and which is used here to denote the probability that an individual alive at exact age x will die within time t thereafter, both x and t being measured in years. The standard actuarial symbol for this probability is ${}_tq_x$ when t is 1 year or less and ${}_tQ_x$ when t is greater than 1 year. The latter notation has been conceded by actuaries to be awkward and unnecessary.⁶ Moreover, a subcommittee designated by the Permanent Committee of the International Congresses of Actuaries to study the revision of the international actuarial notation has gone on record recommending the replacement of the two symbols just mentioned by the one employed here.⁷ The latter symbol has also been widely used, even by actuaries, on the continent of Europe,⁸ and has also appeared in several publications in this country.⁹

Consistency of the tables

Consistency requires that the rates of mortality in the life tables for combinations of classes shall always be intermediate between the rates at the same ages for the component classes. This is true in every case, notwithstanding the fact that the interpolation¹⁰ of the rates of mortality for the combination tables was carried out entirely independently of the corresponding interpolation for the separate classes, except above age

92, where the rates of mortality for separate classes were extrapolated from the data for earlier ages, and those for the various combinations were obtained by a special process in order to insure consistency.

Such consistency as regards the rates of mortality does not, however, guarantee the same kind of consistency in the values of the other life table functions. This would follow if the rates of mortality were obtained by observing a fixed cohort of persons from birth until death, but does not hold when the persons under observation at different ages belong to distinct cohorts, sometimes differing greatly in their race and sex composition. Under these conditions, in fact, such apparent inconsistencies are to be expected, and are not properly regarded as inconsistencies at all. In the life tables in this volume, such situations are few in number and are largely concentrated at the old ages and in the life tables for "other races," and in all these cases the numerical magnitude of the differences involved is small. It may be remarked that such situations have arisen in earlier life tables. For example, in Glover's life table for total males in 1910, the mortality rate is, at every age, intermediate between the corresponding rates for white males and Negro males.¹¹ Nevertheless, the values of l_x at ages 96-98 and d_x at age 55 for total males exceed the corresponding values for both white males and Negro males.¹²

⁶ See *Notation Internationale*, pamphlet issued by the Comité Permanent des Congrès Internationaux d'Actuaires, p. 5, Bruxelles, Février 1939.

⁷ *Op. cit.*, p. 91.

⁸ *Op. cit.*, p. 62.

⁹ See, for example, *American Journal of Hygiene*, vol. 30, No. 2, p. 35 et seq., September 1939; *Record*, American Institute of Actuaries, vol. 32, Part I, No. 65, p. 29 et seq., June 1943.

¹⁰ For a detailed technical description of the process of interpolation, see pp. 122-126.

¹¹ U. S. Bureau of the Census, *United States Life Tables, 1890, 1901, 1910, and 1901-1910*, pp. 58-59, 68-69, 80-81, Government Printing Office, Washington, D. C., 1921.

¹² While it is true that the total males include a small number of males of "other races," this group constituted only 0.16 of 1 percent of the deaths of 1909-1911 at all ages and only 0.17 of 1 percent of the total estimated population, so that this is not likely to be the explanation of the peculiarity noted.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 1.—LIFE TABLE FOR THE TOTAL POPULATION OF THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
0-1	47.10	100,000	4,710	96,058	6,362,494	63.62
1-2	5.21	95,290	496	94,997	6,266,436	65.76
2-3	2.67	94,794	254	94,660	6,171,439	65.10
3-4	1.88	94,540	177	94,448	6,076,779	64.28
4-5	1.51	94,363	143	94,288	5,982,331	63.40
5-6	1.32	94,220	125	94,157	5,888,043	62.49
6-7	1.17	94,095	110	94,041	5,793,886	61.57
7-8	1.05	93,985	98	93,936	5,699,845	60.65
8-9	.96	93,887	91	93,841	5,605,909	59.71
9-10	.91	93,796	86	93,754	5,512,068	58.77
10-11	.90	93,710	84	93,668	5,418,314	57.82
11-12	.92	93,626	86	93,583	5,324,646	56.87
12-13	.97	93,540	91	93,495	5,231,063	55.92
13-14	1.07	93,449	100	93,399	5,137,568	54.98
14-15	1.22	93,349	114	93,292	5,044,169	54.04
15-16	1.39	93,235	130	93,170	4,950,877	53.10
16-17	1.57	93,105	146	93,031	4,857,707	52.17
17-18	1.73	92,959	162	92,878	4,764,676	51.26
18-19	1.88	92,797	174	92,711	4,671,798	50.34
19-20	2.03	92,623	188	92,529	4,579,087	49.44
20-21	2.17	92,435	201	92,334	4,486,558	48.54
21-22	2.30	92,234	212	92,128	4,394,224	47.64
22-23	2.42	92,022	223	91,911	4,302,096	46.75
23-24	2.50	91,799	229	91,684	4,210,185	45.86
24-25	2.56	91,570	235	91,452	4,118,501	44.98
25-26	2.62	91,335	239	91,216	4,027,049	44.09
26-27	2.67	91,096	243	90,974	3,935,833	43.21
27-28	2.75	90,853	250	90,728	3,844,859	42.32
28-29	2.85	90,603	258	90,473	3,754,131	41.44
29-30	2.95	90,345	267	90,212	3,663,658	40.55
30-31	3.07	90,078	276	89,939	3,573,446	39.67
31-32	3.20	89,802	288	89,658	3,483,507	38.79
32-33	3.35	89,514	299	89,365	3,393,849	37.91
33-34	3.51	89,215	313	89,058	3,304,484	37.04
34-35	3.69	88,902	329	88,737	3,215,426	36.17
35-36	3.90	88,573	345	88,401	3,126,689	35.30
36-37	4.12	88,228	363	88,047	3,038,288	34.44
37-38	4.36	87,865	383	87,674	2,950,241	33.58
38-39	4.62	87,482	404	87,279	2,862,567	32.72
39-40	4.91	87,078	428	86,864	2,775,288	31.87
40-41	5.24	86,650	454	86,423	2,688,424	31.03
41-42	5.59	86,196	482	85,955	2,602,001	30.19
42-43	5.99	85,714	513	85,458	2,516,046	29.35
43-44	6.43	85,201	548	84,927	2,430,588	28.53
44-45	6.91	84,653	584	84,361	2,345,661	27.71
45-46	7.44	84,069	626	83,756	2,261,300	26.90
46-47	8.01	83,443	668	83,109	2,177,544	26.10
47-48	8.62	82,775	714	82,418	2,094,435	25.30
48-49	9.28	82,061	761	81,680	2,012,017	24.52
49-50	9.99	81,300	813	80,894	1,930,337	23.74
50-51	10.76	80,487	866	80,054	1,849,443	22.98
51-52	11.59	79,621	923	79,160	1,769,389	22.22
52-53	12.49	78,698	982	78,206	1,690,229	21.48
53-54	13.46	77,716	1,047	77,193	1,612,023	20.74
54-55	14.51	76,669	1,112	76,113	1,534,830	20.02

LIFE TABLES

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TABLE 1.—LIFE TABLE FOR THE TOTAL POPULATION OF THE UNITED STATES: 1939-1941—Continued

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
55-56	15.64	75,557	1,182	74,966	1,458,717	19.31
56-57	16.84	74,375	1,252	73,750	1,383,751	18.60
57-58	18.12	73,123	1,325	72,460	1,310,001	17.92
58-59	19.49	71,798	1,400	71,098	1,237,541	17.24
59-60	20.95	70,398	1,474	69,661	1,166,443	16.57
60-61	22.51	68,924	1,552	68,148	1,096,782	15.91
61-62	24.19	67,372	1,630	66,557	1,028,634	15.27
62-63	26.01	65,742	1,710	64,887	962,077	14.63
63-64	27.97	64,032	1,791	63,137	897,190	14.01
64-65	30.12	62,241	1,875	61,304	834,053	13.40
65-66	32.48	60,366	1,960	59,386	772,749	12.80
66-67	35.09	58,406	2,050	57,381	713,363	12.21
67-68	37.98	56,356	2,140	55,286	655,982	11.64
68-69	41.20	54,216	2,234	53,099	600,696	11.08
69-70	44.77	51,982	2,327	50,818	547,597	10.53
70-71	48.73	49,655	2,420	48,445	496,779	10.00
71-72	53.12	47,235	2,509	45,981	448,334	9.49
72-73	57.98	44,726	2,593	43,430	402,353	9.00
73-74	63.33	42,133	2,668	40,799	358,923	8.52
74-75	69.18	39,465	2,730	38,100	318,124	8.06
75-76	75.54	36,735	2,775	35,347	280,024	7.62
76-77	82.39	33,960	2,798	32,561	244,677	7.20
77-78	89.75	31,162	2,797	29,763	212,116	6.81
78-79	97.61	28,365	2,769	26,981	182,353	6.43
79-80	105.99	25,596	2,713	24,240	155,372	6.07
80-81	114.91	22,883	2,629	21,568	131,132	5.73
81-82	124.38	20,254	2,519	18,995	109,564	5.41
82-83	134.44	17,735	2,385	16,542	90,569	5.11
83-84	145.08	15,350	2,226	14,237	74,027	4.82
84-85	156.25	13,124	2,051	12,099	59,790	4.56
85-86	167.88	11,073	1,859	10,143	47,691	4.31
86-87	179.92	9,214	1,658	8,385	37,548	4.08
87-88	192.29	7,556	1,453	6,830	29,163	3.86
88-89	204.93	6,103	1,250	5,478	22,333	3.66
89-90	217.79	4,853	1,057	4,324	16,855	3.47
90-91	230.81	3,796	876	3,358	12,531	3.30
91-92	243.94	2,920	713	2,563	9,173	3.14
92-93	257.11	2,207	567	1,924	6,610	2.99
93-94	270.31	1,640	443	1,418	4,686	2.86
94-95	283.44	1,197	340	1,027	3,268	2.73
95-96	296.46	857	254	730	2,241	2.61
96-97	309.35	603	186	510	1,511	2.50
97-98	322.10	417	135	350	1,001	2.40
98-99	334.75	282	94	235	651	2.31
99-100	347.36	188	65	155	416	2.21
100-101	360.05	123	45	101	261	2.13
101-102	372.98	78	29	64	160	2.04
102-103	386.34	49	19	39	96	1.96
103-104	400.36	30	12	24	57	1.88
104-105	415.25	18	7	15	33	1.80
105-106	431.17	11	5	8	18	1.72
106-107	448.20	6	3	5	10	1.64
107-108	466.33	3	1	2	5	1.56
108-109	485.39	2	1	2	3	1.48
109-110	505.10	1	1	1	1	1.41

NOTE.—Rates of mortality at ages above 87 are not based on actual statistics at these ages, but have been obtained by mathematical extrapolation from mortality rates at younger ages. Other life table functions at these ages are based on the extrapolated rates of mortality, and may not necessarily represent actual conditions.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 2.—LIFE TABLE FOR TOTAL MALES IN THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
0-1	52.38	100,000	5,238	95,591	6,160,087	61.60
1-2	5.53	94,762	524	94,453	6,064,496	64.00
2-3	2.89	94,238	273	94,093	5,970,043	63.35
3-4	2.01	93,965	189	93,867	5,875,950	62.53
4-5	1.62	93,776	152	93,697	5,782,083	61.66
5-6	1.45	93,624	136	93,556	5,688,386	60.76
6-7	1.30	93,488	121	93,428	5,594,830	59.85
7-8	1.19	93,367	111	93,312	5,501,402	58.92
8-9	1.11	93,256	103	93,204	5,408,090	57.99
9-10	1.06	93,153	99	93,103	5,314,886	57.06
10-11	1.05	93,054	98	93,005	5,221,783	56.12
11-12	1.07	92,956	100	92,906	5,128,778	55.17
12-13	1.13	92,856	105	92,804	5,035,872	54.23
13-14	1.24	92,751	114	92,694	4,943,068	53.29
14-15	1.39	92,637	129	92,572	4,850,374	52.36
15-16	1.57	92,508	146	92,435	4,757,802	51.43
16-17	1.76	92,362	163	92,281	4,665,367	50.51
17-18	1.94	92,199	179	92,110	4,573,086	49.60
18-19	2.11	92,020	194	91,923	4,480,976	48.70
19-20	2.28	91,826	209	91,721	4,389,053	47.80
20-21	2.46	91,617	225	91,504	4,297,332	46.91
21-22	2.61	91,392	239	91,273	4,205,828	46.02
22-23	2.74	91,153	250	91,028	4,114,555	45.14
23-24	2.83	90,903	257	90,774	4,023,527	44.26
24-25	2.88	90,646	261	90,516	3,932,753	43.39
25-26	2.92	90,385	264	90,253	3,842,237	42.51
26-27	2.97	90,121	267	89,988	3,751,984	41.63
27-28	3.04	89,854	273	89,717	3,661,996	40.75
28-29	3.14	89,581	281	89,440	3,572,279	39.88
29-30	3.25	89,300	291	89,155	3,482,839	39.00
30-31	3.38	89,009	300	88,859	3,393,684	38.13
31-32	3.52	88,709	312	88,553	3,304,825	37.25
32-33	3.69	88,397	326	88,233	3,216,272	36.38
33-34	3.88	88,071	341	87,900	3,128,039	35.52
34-35	4.09	87,730	359	87,551	3,040,139	34.66
35-36	4.33	87,371	378	87,182	2,952,588	33.79
36-37	4.59	86,993	399	86,793	2,865,406	32.94
37-38	4.88	86,594	423	86,382	2,778,613	32.09
38-39	5.20	86,171	449	85,946	2,692,231	31.24
39-40	5.56	85,722	476	85,484	2,606,285	30.40
40-41	5.95	85,246	507	84,993	2,520,801	29.57
41-42	6.39	84,739	542	84,467	2,435,808	28.74
42-43	6.87	84,197	578	83,909	2,351,341	27.93
43-44	7.40	83,619	619	83,309	2,267,432	27.12
44-45	7.99	83,000	664	82,668	2,184,123	26.31
45-46	8.63	82,336	710	81,981	2,101,455	25.52
46-47	9.32	81,626	761	81,245	2,019,474	24.74
47-48	10.06	80,865	814	80,458	1,938,229	23.97
48-49	10.86	80,051	869	79,617	1,857,771	23.21
49-50	11.72	79,182	928	78,718	1,778,154	22.46
50-51	12.64	78,254	989	77,759	1,699,436	21.72
51-52	13.64	77,265	1,054	76,738	1,621,677	20.99
52-53	14.72	76,211	1,122	75,650	1,544,939	20.27
53-54	15.90	75,089	1,194	74,492	1,469,289	19.57
54-55	17.16	73,895	1,268	73,261	1,394,797	18.88

LIFE TABLES

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TABLE 2.—LIFE TABLE FOR TOTAL MALES IN THE UNITED STATES: 1939-1941—Continued

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	$\bar{e}_x$
55-56.....	18.50	72,627	1,344	71,955	1,321,536	18.20
56-57.....	19.93	71,283	1,420	70,573	1,249,581	17.53
57-58.....	21.44	69,863	1,498	69,114	1,179,008	16.88
58-59.....	23.02	68,365	1,574	67,579	1,109,894	16.23
59-60.....	24.69	66,791	1,649	65,966	1,042,315	15.61
60-61.....	26.47	65,142	1,724	64,280	976,349	14.99
61-62.....	28.37	63,418	1,800	62,518	912,069	14.38
62-63.....	30.41	61,618	1,873	60,682	849,551	13.79
63-64.....	32.60	59,745	1,948	58,771	788,869	13.20
64-65.....	34.97	57,797	2,021	56,787	730,098	12.63
65-66.....	37.55	55,776	2,094	54,729	673,311	12.07
66-67.....	40.37	53,682	2,167	52,599	618,582	11.52
67-68.....	43.47	51,515	2,239	50,395	565,983	10.99
68-69.....	46.87	49,276	2,310	48,121	515,588	10.46
69-70.....	50.62	46,966	2,378	45,777	467,467	9.95
70-71.....	54.77	44,588	2,442	43,367	421,690	9.46
71-72.....	59.36	42,146	2,502	40,895	378,323	8.98
72-73.....	64.44	39,644	2,555	38,367	337,428	8.51
73-74.....	70.05	37,089	2,598	35,791	299,061	8.06
74-75.....	76.18	34,491	2,627	33,177	263,270	7.63
75-76.....	82.84	31,864	2,640	30,544	230,093	7.22
76-77.....	90.02	29,224	2,631	27,908	199,549	6.83
77-78.....	97.70	26,593	2,598	25,295	171,641	6.45
78-79.....	105.90	23,995	2,541	22,724	146,346	6.10
79-80.....	114.61	21,454	2,459	20,225	123,622	5.76
80-81.....	123.86	18,995	2,353	17,818	103,397	5.44
81-82.....	133.67	16,642	2,224	15,530	85,579	5.14
82-83.....	144.04	14,418	2,077	13,380	70,049	4.86
83-84.....	154.98	12,341	1,912	11,384	56,669	4.59
84-85.....	166.43	10,429	1,736	9,561	45,285	4.34
85-86.....	178.31	8,693	1,550	7,918	35,724	4.11
86-87.....	190.55	7,143	1,361	6,463	27,806	3.89
87-88.....	203.08	5,782	1,174	5,194	21,343	3.69
88-89.....	215.82	4,608	995	4,111	16,149	3.50
89-90.....	228.71	3,613	826	3,200	12,038	3.33
90-91.....	241.68	2,787	674	2,450	8,838	3.17
91-92.....	254.68	2,113	538	1,844	6,388	3.02
92-93.....	267.63	1,575	421	1,364	4,544	2.88
93-94.....	280.66	1,154	324	992	3,180	2.76
94-95.....	293.62	830	244	708	2,188	2.64
95-96.....	306.49	586	179	496	1,480	2.52
96-97.....	319.29	407	130	342	984	2.42
97-98.....	332.09	277	92	231	642	2.32
98-99.....	344.97	185	64	153	411	2.23
99-100.....	358.06	121	43	99	258	2.13
100-101.....	371.53	78	29	63	159	2.05
101-102.....	385.57	49	19	40	96	1.96
102-103.....	400.33	30	12	24	56	1.88
103-104.....	415.94	18	7	14	32	1.79
104-105.....	432.43	11	5	8	18	1.71
105-106.....	449.65	6	3	5	10	1.64
106-107.....	467.23	3	1	2	5	1.57
107-108.....	484.46	2	1	2	3	1.51
108-109.....	500.29	1	1	1	1	1.46

NOTE.—Rates of mortality at ages above 92 are not based on actual statistics at these ages, but have been obtained by mathematical extrapolation from mortality rates at younger ages. Other life table functions at these ages are based on the extrapolated rates of mortality, and may not necessarily represent actual conditions.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 3.—LIFE TABLE FOR TOTAL FEMALES IN THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	$1,000q_x$	l_x	d_x	L_x	T_x	e_x
0-1	41.52	100,000	4,152	96,549	6,588,801	65.89
1-2	4.89	95,848	469	95,571	6,492,252	67.73
2-3	2.44	95,379	232	95,256	6,396,681	67.07
3-4	1.74	95,147	166	95,061	6,301,425	66.23
4-5	1.40	94,981	133	94,912	6,206,364	65.34
5-6	1.20	94,848	114	94,791	6,111,452	64.43
6-7	1.03	94,734	97	94,685	6,016,661	63.51
7-8	.90	94,637	86	94,594	5,921,976	62.58
8-9	.82	94,551	77	94,513	5,827,382	61.63
9-10	.76	94,474	72	94,438	5,732,869	60.68
10-11	.75	94,402	70	94,367	5,638,431	59.73
11-12	.76	94,332	72	94,296	5,544,064	58.77
12-13	.81	94,260	76	94,222	5,449,768	57.82
13-14	.90	94,184	85	94,141	5,355,546	56.86
14-15	1.04	94,099	99	94,049	5,261,405	55.91
15-16	1.21	94,000	113	93,944	5,167,356	54.97
16-17	1.38	93,887	130	93,822	5,073,412	54.04
17-18	1.53	93,757	143	93,686	4,979,590	53.11
18-19	1.65	93,614	155	93,536	4,885,904	52.19
19-20	1.78	93,459	166	93,377	4,792,368	51.28
20-21	1.90	93,293	177	93,204	4,698,991	50.37
21-22	2.01	93,116	186	93,024	4,605,787	49.46
22-23	2.11	92,930	196	92,831	4,512,763	48.56
23-24	2.19	92,734	203	92,633	4,419,932	47.66
24-25	2.26	92,531	209	92,427	4,327,299	46.77
25-26	2.32	92,322	214	92,214	4,234,872	45.87
26-27	2.39	92,108	221	91,998	4,142,658	44.98
27-28	2.47	91,887	227	91,774	4,050,660	44.08
28-29	2.57	91,660	235	91,542	3,958,886	43.19
29-30	2.66	91,425	243	91,304	3,867,344	42.30
30-31	2.77	91,182	253	91,055	3,776,040	41.41
31-32	2.89	90,929	262	90,798	3,684,985	40.53
32-33	3.01	90,667	274	90,530	3,594,187	39.64
33-34	3.15	90,393	285	90,251	3,503,657	38.76
34-35	3.31	90,108	298	89,959	3,413,406	37.88
35-36	3.47	89,810	311	89,655	3,323,447	37.01
36-37	3.65	89,499	327	89,335	3,233,792	36.13
37-38	3.84	89,172	342	89,001	3,144,457	35.26
38-39	4.05	88,830	360	88,650	3,055,456	34.40
39-40	4.27	88,470	378	88,281	2,966,806	33.53
40-41	4.52	88,092	398	87,893	2,878,525	32.68
41-42	4.79	87,694	420	87,484	2,790,632	31.82
42-43	5.10	87,274	445	87,052	2,703,148	30.97
43-44	5.43	86,829	471	86,593	2,616,096	30.13
44-45	5.80	86,358	502	86,107	2,529,503	29.29
45-46	6.21	85,856	533	85,590	2,443,396	28.46
46-47	6.65	85,323	567	85,040	2,357,806	27.63
47-48	7.12	84,756	604	84,454	2,272,766	26.82
48-49	7.63	84,152	641	83,831	2,188,312	26.00
49-50	8.17	83,511	683	83,169	2,104,481	25.20
50-51	8.76	82,828	725	82,466	2,021,312	24.40
51-52	9.40	82,103	772	81,717	1,938,846	23.61
52-53	10.09	81,331	820	80,921	1,857,129	22.83
53-54	10.85	80,511	874	80,074	1,776,208	22.06
54-55	11.67	79,637	929	79,173	1,696,134	21.30

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TABLE 3.—LIFE TABLE FOR TOTAL FEMALES IN THE UNITED STATES: 1939-1941—Continued

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
55-56	12.57	78,708	989	78,213	1,616,961	20.54
56-57	13.54	77,719	1,052	77,193	1,538,748	19.80
57-58	14.60	76,667	1,120	76,107	1,461,555	19.06
58-59	15.75	75,547	1,190	74,952	1,385,448	18.34
59-60	17.00	74,357	1,264	73,726	1,310,496	17.62
60-61	18.37	73,093	1,342	72,421	1,236,770	16.92
61-62	19.85	71,751	1,425	71,039	1,164,349	16.23
62-63	21.47	70,326	1,510	69,571	1,093,310	15.55
63-64	23.24	68,816	1,599	68,016	1,023,739	14.88
64-65	25.19	67,217	1,694	66,370	955,723	14.22
65-66	27.36	65,523	1,792	64,627	889,353	13.57
66-67	29.78	63,731	1,899	62,782	824,726	12.94
67-68	32.50	61,832	2,009	60,828	761,944	12.32
68-69	35.54	59,823	2,127	58,759	701,116	11.72
69-70	38.95	57,696	2,247	56,573	642,357	11.13
70-71	42.74	55,449	2,370	54,264	585,784	10.56
71-72	46.96	53,079	2,493	51,833	531,520	10.01
72-73	51.63	50,586	2,612	49,280	479,687	9.48
73-74	56.79	47,974	2,724	46,612	430,407	8.97
74-75	62.43	45,250	2,825	43,838	383,795	8.48
75-76	68.56	42,425	2,909	40,971	339,957	8.01
76-77	75.19	39,516	2,971	38,031	298,986	7.57
77-78	82.33	36,545	3,009	35,041	260,955	7.14
78-79	89.97	33,536	3,017	32,027	225,914	6.74
79-80	98.14	30,519	2,995	29,022	193,887	6.35
80-81	106.87	27,524	2,942	26,053	164,865	5.99
81-82	116.18	24,582	2,856	23,154	138,812	5.65
82-83	126.09	21,726	2,739	20,357	115,658	5.32
83-84	136.62	18,987	2,594	17,690	95,301	5.02
84-85	147.72	16,393	2,421	15,182	77,611	4.73
85-86	159.32	13,972	2,226	12,859	62,429	4.47
86-87	171.38	11,746	2,013	10,739	49,570	4.22
87-88	183.83	9,733	1,790	8,838	38,831	3.99
88-89	196.61	7,943	1,561	7,163	29,993	3.78
89-90	209.67	6,382	1,338	5,712	22,830	3.58
90-91	222.96	5,044	1,125	4,482	17,118	3.39
91-92	236.44	3,919	927	3,456	12,636	3.22
92-93	250.05	2,992	748	2,618	9,180	3.07
93-94	263.53	2,244	591	1,948	6,562	2.92
94-95	276.92	1,653	458	1,424	4,614	2.79
95-96	290.19	1,195	347	1,022	3,190	2.67
96-97	303.27	848	257	720	2,168	2.56
97-98	316.13	591	187	497	1,448	2.45
98-99	328.79	404	133	338	951	2.35
99-100	341.27	271	92	225	613	2.26
100-101	353.68	179	63	147	388	2.17
101-102	366.19	116	43	94	241	2.09
102-103	379.03	73	28	60	147	2.00
103-104	392.49	45	17	36	87	1.92
104-105	406.91	28	12	22	51	1.83
105-106	422.58	16	7	13	29	1.75
106-107	439.78	9	4	8	16	1.67
107-108	458.69	5	2	4	8	1.58
108-109	479.41	3	2	2	4	1.49
109-110	501.93	1	0	1	2	1.41
110-111	526.10	1	1	1	1	1.33

NOTE.—Rates of mortality at ages above 87 are not based on actual statistics at these ages, but have been obtained by mathematical extrapolation from mortality rates younger ages. Other life table functions at these ages are based on the extrapolated rates of mortality, and may not necessarily represent actual conditions.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 4.—LIFE TABLE FOR TOTAL WHITES IN THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	$1,000q_x$	l_x	d_x	L_x	T_x	e_x
0-1	43.15	100,000	4,315	96,354	6,492,419	64.92
1-2	4.60	95,685	440	95,425	6,396,065	66.84
2-3	2.43	95,245	231	95,123	6,300,640	66.15
3-4	1.76	95,014	167	94,927	6,205,517	65.31
4-5	1.41	94,847	134	94,777	6,110,590	64.43
5-6	1.24	94,713	118	94,654	6,015,813	63.52
6-7	1.10	94,595	104	94,543	5,921,159	62.59
7-8	1.00	94,491	94	94,444	5,826,616	61.66
8-9	.92	94,397	87	94,353	5,732,172	60.72
9-10	.87	94,310	82	94,269	5,637,819	59.78
10-11	.85	94,228	81	94,187	5,543,550	58.83
11-12	.86	94,147	81	94,107	5,449,363	57.88
12-13	.89	94,066	83	94,025	5,355,256	56.93
13-14	.96	93,983	91	93,937	5,261,231	55.98
14-15	1.07	93,892	100	93,842	5,167,294	55.03
15-16	1.20	93,792	113	93,735	5,073,452	54.09
16-17	1.33	93,679	124	93,617	4,979,717	53.16
17-18	1.45	93,555	136	93,487	4,886,100	52.23
18-19	1.56	93,419	146	93,347	4,792,613	51.30
19-20	1.67	93,273	156	93,195	4,699,266	50.38
20-21	1.78	93,117	166	93,034	4,606,071	49.47
21-22	1.88	92,951	175	92,864	4,513,037	48.55
22-23	1.97	92,776	182	92,685	4,420,173	47.64
23-24	2.03	92,594	189	92,499	4,327,488	46.74
24-25	2.08	92,405	192	92,310	4,234,989	45.83
25-26	2.12	92,213	195	92,115	4,142,679	44.92
26-27	2.16	92,018	199	91,919	4,050,564	44.02
27-28	2.23	91,819	204	91,717	3,958,645	43.11
28-29	2.30	91,615	212	91,509	3,866,928	42.21
29-30	2.39	91,403	218	91,294	3,775,419	41.31
30-31	2.49	91,185	228	91,071	3,684,125	40.40
31-32	2.60	90,957	236	90,839	3,593,054	39.50
32-33	2.73	90,721	248	90,597	3,502,215	38.60
33-34	2.87	90,473	259	90,343	3,411,618	37.71
34-35	3.03	90,214	273	90,077	3,321,275	36.82
35-36	3.20	89,941	288	89,797	3,231,198	35.93
36-37	3.40	89,653	305	89,500	3,141,401	35.04
37-38	3.61	89,348	322	89,187	3,051,901	34.16
38-39	3.85	89,026	343	88,855	2,962,714	33.28
39-40	4.11	88,683	365	88,501	2,873,859	32.41
40-41	4.41	88,318	389	88,123	2,785,358	31.54
41-42	4.74	87,929	416	87,721	2,697,235	30.68
42-43	5.11	87,513	447	87,289	2,609,514	29.82
43-44	5.52	87,066	480	86,826	2,522,225	28.97
44-45	5.97	86,586	517	86,327	2,435,399	28.13
45-46	6.46	86,069	557	85,791	2,349,072	27.29
46-47	7.00	85,512	598	85,213	2,263,281	26.47
47-48	7.59	84,914	645	84,591	2,178,068	25.65
48-49	8.22	84,269	693	83,923	2,093,477	24.84
49-50	8.90	83,576	743	83,204	2,009,554	24.04
50-51	9.64	82,833	799	82,434	1,926,350	23.26
51-52	10.45	82,034	857	81,605	1,843,916	22.48
52-53	11.32	81,177	919	80,717	1,762,311	21.71
53-54	12.28	80,258	985	79,766	1,681,594	20.95
54-55	13.31	79,273	1,055	78,745	1,601,828	20.21

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TABLE 4.—LIFE TABLE FOR TOTAL WHITES IN THE UNITED STATES: 1939-1941—Continued

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	$1,000q_x$	l_x	d_x	L_x	T_x	e_x
55-56.....	14.43	78,218	1,129	77,653	1,523,083	19.47
56-57.....	15.63	77,089	1,205	76,487	1,445,430	18.75
57-58.....	16.92	75,884	1,284	75,242	1,368,943	18.04
58-59.....	18.31	74,600	1,365	73,918	1,293,701	17.34
59-60.....	19.79	73,235	1,450	72,510	1,219,783	16.66
60-61.....	21.40	71,785	1,536	71,017	1,147,273	15.98
61-62.....	23.12	70,249	1,624	69,437	1,076,256	15.32
62-63.....	24.99	68,625	1,715	67,767	1,006,819	14.67
63-64.....	27.01	66,910	1,807	66,007	939,052	14.03
64-65.....	29.22	65,103	1,902	64,151	873,045	13.41
65-66.....	31.64	63,201	2,000	62,201	808,894	12.80
66-67.....	34.33	61,201	2,101	60,150	746,693	12.20
67-68.....	37.31	59,100	2,206	57,997	686,543	11.62
68-69.....	40.63	56,894	2,311	55,739	628,546	11.05
69-70.....	44.31	54,583	2,418	53,374	572,807	10.49
70-71.....	48.39	52,165	2,524	50,903	519,433	9.96
71-72.....	52.90	49,641	2,626	48,328	468,530	9.44
72-73.....	57.88	47,015	2,721	45,654	420,202	8.94
73-74.....	63.36	44,294	2,807	42,890	374,548	8.46
74-75.....	69.34	41,487	2,877	40,049	331,658	7.99
75-76.....	75.83	38,610	2,927	37,146	291,609	7.55
76-77.....	82.82	35,683	2,955	34,206	254,463	7.13
77-78.....	90.31	32,728	2,956	31,249	220,257	6.73
78-79.....	98.32	29,772	2,927	28,309	189,008	6.35
79-80.....	106.87	26,845	2,869	25,410	160,699	5.99
80-81.....	115.99	23,976	2,781	22,585	135,289	5.64
81-82.....	125.73	21,195	2,665	19,863	112,704	5.32
82-83.....	136.12	18,530	2,522	17,268	92,841	5.01
83-84.....	147.17	16,008	2,356	14,830	75,573	4.72
84-85.....	158.85	13,652	2,169	12,568	60,743	4.45
85-86.....	171.09	11,483	1,964	10,500	48,175	4.20
86-87.....	183.84	9,519	1,750	8,644	37,675	3.96
87-88.....	197.03	7,769	1,531	7,003	29,031	3.74
88-89.....	210.61	6,238	1,314	5,581	22,028	3.53
89-90.....	224.53	4,924	1,105	4,372	16,447	3.34
90-91.....	238.74	3,819	912	3,363	12,075	3.16
91-92.....	253.20	2,907	736	2,539	8,712	3.00
92-93.....	267.84	2,171	582	1,880	6,173	2.84
93-94.....	282.74	1,589	449	1,364	4,293	2.70
94-95.....	297.77	1,140	339	971	2,929	2.57
95-96.....	312.88	801	251	675	1,958	2.45
96-97.....	328.03	550	180	460	1,283	2.33
97-98.....	343.18	370	127	306	823	2.23
98-99.....	358.27	243	87	199	517	2.13
99-100.....	373.27	156	58	127	318	2.04
100-101.....	388.11	98	38	79	191	1.95
101-102.....	402.76	60	24	48	112	1.88
102-103.....	417.14	36	15	28	64	1.81
103-104.....	431.21	21	9	16	36	1.74
104-105.....	444.89	12	5	9	20	1.68
105-106.....	458.10	7	3	5	11	1.62
106-107.....	470.78	4	2	3	6	1.57
107-108.....	482.81	2	1	2	3	1.53
108-109.....	494.08	1	1	1	1	1.48

NOTE.—Rates of mortality at ages above 92 are not based on actual statistics at these ages, but have been obtained by mathematical extrapolation from mortality rates at younger ages. Other life table functions at these ages are based on the extrapolated rates of mortality, and may not necessarily represent actual conditions.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 5.—LIFE TABLE FOR WHITE MALES IN THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
0-1	48.12	100,000	4,812	95,913	6,281,188	62.81
1-2	4.87	95,188	464	94,914	6,185,275	64.98
2-3	2.65	94,724	250	94,592	6,090,361	64.30
3-4	1.90	94,474	179	94,381	5,995,769	63.46
4-5	1.53	94,295	145	94,219	5,901,388	62.58
5-6	1.38	94,150	130	94,085	5,807,169	61.68
6-7	1.24	94,020	116	93,962	5,713,084	60.76
7-8	1.14	93,904	108	93,850	5,619,122	59.84
8-9	1.06	93,796	99	93,747	5,525,272	58.91
9-10	1.02	93,697	96	93,649	5,431,525	57.97
10-11	1.00	93,601	93	93,554	5,337,876	57.03
11-12	1.01	93,508	95	93,460	5,244,322	56.08
12-13	1.06	93,413	99	93,364	5,150,862	55.14
13-14	1.14	93,314	106	93,261	5,057,498	54.20
14-15	1.27	93,208	119	93,148	4,964,237	53.26
15-16	1.43	93,089	133	93,023	4,871,089	52.33
16-17	1.58	92,956	147	92,882	4,778,066	51.40
17-18	1.72	92,809	160	92,729	4,685,184	50.48
18-19	1.86	92,649	172	92,563	4,592,455	49.57
19-20	1.99	92,477	184	92,385	4,499,892	48.66
20-21	2.12	92,293	195	92,195	4,407,507	47.76
21-22	2.23	92,098	205	91,996	4,315,312	46.86
22-23	2.32	91,893	214	91,785	4,223,316	45.96
23-24	2.38	91,679	218	91,571	4,131,531	45.07
24-25	2.41	91,461	220	91,351	4,039,960	44.17
25-26	2.43	91,241	222	91,130	3,948,609	43.28
26-27	2.45	91,019	223	90,908	3,857,479	42.38
27-28	2.51	90,796	228	90,682	3,766,571	41.48
28-29	2.59	90,568	234	90,451	3,675,889	40.59
29-30	2.68	90,334	242	90,212	3,585,438	39.69
30-31	2.79	90,092	251	89,967	3,495,226	38.80
31-32	2.91	89,841	262	89,709	3,405,259	37.90
32-33	3.06	89,579	274	89,443	3,315,550	37.01
33-34	3.23	89,305	288	89,161	3,226,107	36.12
34-35	3.42	89,017	304	88,865	3,136,946	35.24
35-36	3.63	88,713	322	88,552	3,048,081	34.36
36-37	3.87	88,391	342	88,220	2,959,529	33.48
37-38	4.14	88,049	364	87,867	2,871,309	32.61
38-39	4.43	87,685	389	87,490	2,783,442	31.74
39-40	4.76	87,296	416	87,088	2,695,952	30.88
40-41	5.13	86,880	446	86,657	2,608,864	30.03
41-42	5.54	86,434	479	86,195	2,522,207	29.18
42-43	6.00	85,955	515	85,698	2,436,012	28.34
43-44	6.50	85,440	555	85,162	2,350,314	27.51
44-45	7.06	84,885	600	84,585	2,265,152	26.69
45-46	7.66	84,285	646	83,962	2,180,567	25.87
46-47	8.33	83,639	696	83,292	2,096,605	25.07
47-48	9.04	82,943	750	82,568	2,013,313	24.27
48-49	9.81	82,193	806	81,790	1,930,745	23.49
49-50	10.64	81,387	866	80,954	1,848,955	22.72
50-51	11.55	80,521	930	80,056	1,768,001	21.96
51-52	12.53	79,591	997	79,092	1,687,945	21.21
52-53	13.60	78,594	1,069	78,059	1,608,853	20.47
53-54	14.76	77,525	1,145	76,953	1,530,794	19.75
54-55	16.02	76,380	1,224	75,768	1,453,841	19.03

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TABLE 5.—LIFE TABLE FOR WHITE MALES IN THE UNITED STATES: 1939-1941—Continued

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	1,000 q_x	l_x	d_x	L_x	T_x	e_x
55-56.....	17.37	75,156	1,305	74,504	1,378,073	18.34
56-57.....	18.81	73,851	1,390	73,156	1,303,569	17.65
57-58.....	20.34	72,461	1,473	71,724	1,230,413	16.98
58-59.....	21.95	70,988	1,558	70,209	1,158,689	16.32
59-60.....	23.66	69,430	1,643	68,609	1,088,480	15.68
60-61.....	25.48	67,787	1,727	66,923	1,019,871	15.05
61-62.....	27.43	66,060	1,813	65,153	952,948	14.43
62-63.....	29.52	64,247	1,896	63,299	887,795	13.82
63-64.....	31.77	62,351	1,981	61,361	824,496	13.22
64-65.....	34.20	60,370	2,065	59,337	763,135	12.64
65-66.....	36.85	58,305	2,148	57,232	703,798	12.07
66-67.....	39.75	56,157	2,232	55,041	646,566	11.51
67-68.....	42.93	53,925	2,315	52,767	591,525	10.97
68-69.....	46.43	51,610	2,396	50,412	538,758	10.44
69-70.....	50.28	49,214	2,475	47,976	488,346	9.92
70-71.....	54.54	46,739	2,549	45,465	440,370	9.42
71-72.....	59.24	44,190	2,618	42,881	394,905	8.94
72-73.....	64.43	41,572	2,678	40,233	352,024	8.47
73-74.....	70.14	38,894	2,728	37,530	311,791	8.02
74-75.....	76.37	36,166	2,762	34,784	274,261	7.58
75-76.....	83.13	33,404	2,777	32,016	239,477	7.17
76-77.....	90.40	30,627	2,769	29,243	207,461	6.77
77-78.....	98.18	27,858	2,735	26,490	178,218	6.40
78-79.....	106.47	25,123	2,675	23,786	151,728	6.04
79-80.....	115.30	22,448	2,588	21,155	127,942	5.70
80-81.....	124.71	19,860	2,477	18,621	106,787	5.38
81-82.....	134.72	17,383	2,341	16,213	88,166	5.07
82-83.....	145.37	15,042	2,187	13,948	71,953	4.78
83-84.....	156.68	12,855	2,014	11,848	58,005	4.51
84-85.....	168.59	10,841	1,828	9,927	46,157	4.26
85-86.....	181.04	9,013	1,631	8,198	36,230	4.02
86-87.....	193.95	7,382	1,432	6,665	28,032	3.80
87-88.....	207.27	5,950	1,233	5,334	21,367	3.59
88-89.....	220.91	4,717	1,042	4,195	16,033	3.40
89-90.....	234.82	3,675	863	3,244	11,838	3.22
90-91.....	248.94	2,812	700	2,461	8,594	3.06
91-92.....	263.22	2,112	556	1,834	6,133	2.90
92-93.....	277.60	1,556	432	1,340	4,299	2.76
93-94.....	292.02	1,124	328	960	2,959	2.63
94-95.....	306.42	796	244	674	1,999	2.51
95-96.....	320.76	552	177	464	1,325	2.40
96-97.....	334.96	375	126	312	861	2.30
97-98.....	348.98	249	87	205	549	2.20
98-99.....	362.75	162	59	133	344	2.12
99-100.....	376.23	103	38	84	211	2.04
100-101.....	389.35	65	26	52	127	1.96
101-102.....	402.05	39	15	32	75	1.90
102-103.....	414.29	24	10	18	43	1.84
103-104.....	425.99	14	6	11	25	1.78
104-105.....	437.12	8	4	6	14	1.73
105-106.....	447.60	4	2	4	8	1.68
106-107.....	457.38	2	1	2	4	1.64
107-108.....	466.40	1	0	1	2	1.61
108-109.....	474.62	1	1	1	1	1.57

NOTE.—Rates of mortality at ages above 92 are not based on actual statistics at these ages, but have been obtained by mathematical extrapolation from mortality rates at younger ages. Other life table functions at these ages are based on the extrapolated rates of mortality, and may not necessarily represent actual conditions.

UNITED STATES LIFE TABLES AND ACTUARIAL TABLES

TABLE 6.—LIFE TABLE FOR WHITE FEMALES IN THE UNITED STATES: 1939-1941

YEAR OF AGE	MORTALITY RATE	OF 100,000 BORN ALIVE		STATIONARY POPULATION		AVERAGE FUTURE LIFETIME
Period of life between two exact ages stated	Number dying per 1,000 alive at beginning of year of age	Number living at beginning of year of age	Number dying during year of age	In year of age	In year of age and all later years	Average number of years of life remaining at beginning of year of age
(1)	(2)	(3)	(4)	(5)	(6)	(7)
x to $x+1$	$1,000q_x$	l_x	d_x	L_x	T_x	e_x
0-1	37.89	100,000	3,789	96,822	6,728,965	67.29
1-2	4.32	96,211	415	95,966	6,632,143	68.93
2-3	2.20	95,796	211	95,684	6,536,177	68.23
3-4	1.61	95,585	154	95,505	6,440,493	67.38
4-5	1.28	95,431	122	95,367	6,344,988	66.49
5-6	1.10	95,309	106	95,256	6,249,621	65.57
6-7	.96	95,203	91	95,158	6,154,365	64.64
7-8	.85	95,112	80	95,072	6,059,207	63.71
8-9	.77	95,032	74	94,995	5,964,135	62.76
9-10	.72	94,958	68	94,924	5,869,140	61.81
10-11	.70	94,890	66	94,857	5,774,216	60.85
11-12	.70	94,824	66	94,791	5,679,359	59.89
12-13	.72	94,758	69	94,723	5,584,568	58.94
13-14	.77	94,689	73	94,653	5,489,845	57.98
14-15	.86	94,616	82	94,575	5,395,192	57.02
15-16	.96	94,534	91	94,489	5,300,617	56.07
16-17	1.07	94,443	101	94,392	5,206,128	55.12
17-18	1.17	94,342	111	94,287	5,111,736	54.18
18-19	1.26	94,231	119	94,172	5,017,449	53.25
19-20	1.36	94,112	128	94,048	4,923,277	52.31
20-21	1.45	93,984	136	93,916	4,829,229	51.38
21-22	1.54	93,848	145	93,776	4,735,313	50.46
22-23	1.62	93,703	152	93,627	4,641,537	49.53
23-24	1.70	93,551	159	93,472	4,547,910	48.61
24-25	1.76	93,392	164	93,310	4,454,438	47.70
25-26	1.82	93,228	169	93,144	4,361,128	46.78
26-27	1.88	93,059	175	92,972	4,267,984	45.86
27-28	1.95	92,884	181	92,793	4,175,012	44.95
28-29	2.03	92,703	188	92,610	4,082,219	44.04
29-30	2.11	92,515	195	92,417	3,989,609	43.12
30-31	2.20	92,320	204	92,218	3,897,192	42.21
31-32	2.30	92,116	212	92,010	3,804,974	41.31
32-33	2.40	91,904	220	91,794	3,712,964	40.40
33-34	2.52	91,684	231	91,568	3,621,170	39.50
34-35	2.64	91,453	242	91,332	3,529,602	38.59
35-36	2.78	91,211	253	91,085	3,438,270	37.70
36-37	2.92	90,958	266	90,825	3,347,185	36.80
37-38	3.09	90,692	280	90,552	3,256,360	35.91
38-39	3.26	90,412	295	90,265	3,165,808	35.02
39-40	3.46	90,117	312	89,961	3,075,543	34.13
40-41	3.68	89,805	330	89,640	2,985,582	33.25
41-42	3.93	89,475	352	89,299	2,895,942	32.37
42-43	4.20	89,123	374	88,936	2,806,643	31.49
43-44	4.51	88,749	400	88,549	2,717,707	30.62
44-45	4.85	88,349	429	88,134	2,629,158	29.76
45-46	5.23	87,920	460	87,690	2,541,024	28.90
46-47	5.64	87,460	493	87,214	2,453,334	28.05
47-48	6.08	86,967	528	86,703	2,366,120	27.21
48-49	6.55	86,439	566	86,156	2,279,417	26.37
49-50	7.06	85,873	606	85,570	2,193,261	25.54
50-51	7.62	85,267	650	84,942	2,107,691	24.72
51-52	8.22	84,617	695	84,269	2,022,749	23.90
52-53	8.88	83,922	746	83,549	1,938,480	23.10
53-54	9.61	83,176	799	82,777	1,854,931	22.30
54-55	10.40	82,377	857	81,948	1,772,154	21.51